

Chinese SkillSpan Manual Annotation Handbook

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Chinese is the annotation source. English glosses explain the meaning of the Chinese examples; they are not independently annotated English examples. Labels, highlighted boundaries, and character offsets refer only to the Chinese text.

Preface and instructions for use

This handbook guides the annotation of competence spans in Chinese job advertisements for Chinese SkillSpan. It is intended for annotators, reviewers, and researchers using the dataset. The task is to identify source-text spans expressing applicants' required language proficiency (L), knowledge (K), occupational skills (S), and transversal competences (T), and record each span and its type. The handbook explains scope, span boundaries, classification, and the handling of difficult cases, with examples for reference.

Conceptual sources and annotation principles

ESCO, the European classification of skills, competences, qualifications and occupations, and the European Qualifications Framework (EQF) provide conceptual reference points for knowledge and skills. [ESCO14] [EQF] The research team specifies additional annotation rules for Chinese job advertisements: education and non-language occupational qualifications are K; domain names in explicit industry-background requirements are broad K; tool names are K or S according to the requirement expressed. These are project rules, not labels assigned to individual Chinese expressions by ESCO.

Select each entity as a continuous substring of the source, keeping it as short as possible without losing the requirement. Do not rewrite the text or insert words. Final annotations are flat and non-overlapping. The same term may receive different types in different contexts; introductory expressions such as 熟悉 (be familiar with) and 掌握 (have a command of) do not determine the type. Use only the current sentence, including its component clauses. Do not draw on neighboring sentences, the job title, or the full advertisement to supply missing meaning.

Scope of these revisions: review the proposed rules before adopting them for the next annotation batch, and record the approved version when they are used. Historical labels, predictions, and scores remain associated with their original versions. Full cases not yet reviewed against the revised criteria are identified beside the examples; they are not validated reference answers for the new rules.

How to read and consult the handbook

Start with the four label definitions, then read the scope and boundary rules. In practice, ask in order: Is the expression in scope? Which source characters should be selected? Which type applies? Consult the linked cases when uncertain, and submit unresolved cases for review.

A span is the continuous source substring selected for annotation; its boundaries are its start and end positions. Confirmed empty means the text has been checked and contains no target entity. Unresolved means the evidence is insufficient or a disagreement remains. Record these states separately. A provisional S is a temporary entry while the type is undecided: retain the candidate types and reasons and submit the record for review. It does not establish an occupational-skill label. Scope and boundaries must already be settled.

When annotators disagree, record the alternatives and reasons and submit them for review. The reviewer checks the source and rules, confirms the final annotation, and records the rationale. If no decision can be reached, retain the pending-review status. This review and confirmation process is commonly called adjudication. Record model review separately from human confirmation. Keep the historical field name `adjudication_required`; it indicates that review is required.

R08–R23 identify supplementary rules; EX001–EX124 identify Appendix A entries. Numbers such as case 128 retain the original record identifiers. Cases explain rules and annotation rationales. Their counts refer to reference entries, not independently evaluated samples. Markers such as [Z22] link to the references at the end; [Project rule] denotes a rule established for this project, not an external reference.

Data terminology: Gold means reference labels produced through the applicable human-confirmation procedure; independence and training or evaluation use must be stated separately. Silver-plus consists of model-generated labels that have been checked or revised under the project procedure for use in training. Sampling some records for review does not make the whole set Gold. IAA is inter-annotator agreement; report the metric and comparison conditions. JSONL stores one record per line. A negative example here is a record confirmed to contain no target entity, not an undecided record.

Four labels and their evaluation grouping

Human annotations retain L, K, S, and T. In two-class evaluation, K and L count as KNOWLEDGE, while S and T count as SKILL; the original human labels remain unchanged. This grouping is the project's rule for comparison with the two-class SkillSpan scheme. [Z22] [Project rule]

Related examples: [EX001](#)–[EX004](#) (Appendix A)

Concepts and annotation rules: ESCO and EQF provide reference definitions of knowledge and skills. The project uses these to define annotation procedures for Chinese job advertisements, assigning eligible source spans to L, K, S, or T. Education and non-language qualifications are K; domain names in explicit industry-background requirements are broad K; tool names are S or K according to the requirement. These decisions are project rules, not word-by-word labels supplied by ESCO. [ESCO14] [EQF] [Project rule]

SkillSpan [\[Z22\]](#) provides reference guidance on span selection, knowledge/skill examples, and two-class evaluation. The table below defines this project's four labels.

K Knowledge

Related ESCO category: Knowledge

ESCO/EQF definition: “Knowledge means the outcome of the assimilation of information through learning. Knowledge is the body of facts, principles, theories and practices that is related to a field of work or study.” For annotation, consider the facts, principles, theories, and organized knowledge of a domain. The presence or absence of a verb alone does not determine K. [\[EQF\]](#) [\[ESCO14\]](#)

Chinese job-text annotation: domain facts, principles, and theories; domain names in explicit industry-background requirements (broad K under this project); and education, degree, and non-language qualification names (K under project rules).

Examples: 本科及以上学历 (bachelor's degree or higher), 计算机专业 (a computer science major), ISO 27001, PMP.

In two-class evaluation, K counts as KNOWLEDGE. [Project rule]

S Occupational skills

Related ESCO category: Skills

ESCO/EQF definition: “Skill means the ability to apply knowledge and use know-how to complete tasks and solve problems.”

Chinese job-text annotation: applying knowledge through tools, methods, or specific occupational activities.

Examples: 维护 (maintenance), 测试 (testing), 接口对接 (interface integration), and shell in an occupational-use context.

In two-class evaluation, S counts as SKILL. [Project rule]

T Transversal competences

Related ESCO category: Transversal skills

Definition: T covers general abilities and dispositions usable across work, learning, and everyday settings, including communication, cooperation, and self-management. ESCO's social and communication categories include communicating and leading others; report facts is under T4.1. T4.1 is an ESCO subcategory, not an additional project label; annotate T only. [\[ESCO-T\]](#)

Chinese job-text annotation: general communication, cooperation, self-management, and related dispositions are assessed as T. Their placement in a job-duty sentence does not by itself make them S.

Examples: 沟通能力 (communication skills), 沟通管理 (communication management), 客户汇报 (reporting to clients), 责任心 (sense of responsibility).

In two-class evaluation, T counts as SKILL. [Project rule]

L Language proficiency

Related ESCO category: Language skills and knowledge

Classification: ESCO lists Language skills and knowledge as a separate category, L, alongside knowledge, skills, and transversal skills. This category includes knowledge concepts such as language names as well as listening, speaking, reading, and writing skills. [ESCO-L]

Chinese job-text annotation: natural-language names, proficiency, listening/speaking/reading/writing abilities, and language examinations or certificates belong to L. For communication expressions, distinguish a requirement for language proficiency from a requirement to contact or coordinate with people; the latter is assessed as T.

Examples: 英语 (English), 英语六级 (College English Test Band 6), CET-6, 日语 N2 (Japanese proficiency at N2 level), 英文阅读能力 (English reading ability). 英语六级 and CET-6 are alternative forms of a language qualification and both receive L. Preserve whichever form occurs in the source. Language here means natural language, not programming languages such as Java or Python.

In two-class evaluation, L counts as KNOWLEDGE. [Project rule]

Conceptual and methodological references

- Four branches of the skills pillar: skill_main (https://esco.ec.europa.eu/en/classification/skill_main) · Skills pillar (<https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/skills-pillar>) [ESCO-L] [ESCO14]
- Knowledge: escopedia/knowledge (<https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/knowledge>) [EQF]
- Skill: escopedia/skill (<https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/skill>) [EQF]
- Transversal / report facts: TSC 2022 (<https://esco.ec.europa.eu/en/about-esco/publications/publication/towards-structured-and-consistent-terminology-transversal>) · report facts (<https://esco.ec.europa.eu/en/classification/skills?uri=http://data.europa.eu/esco/skill/be6ab363-3de1-427f-a8ef-85d5b0250822>) [ESCO-T]
- Qualifications: ESCO Qualifications (<https://esco.ec.europa.eu/en/classification/qualifications>). [ESCO-Q]

Order of decisions when resolving ambiguity

Related examples: [EX005](#) (Appendix A)

Determine the type from the current context and the four operational definitions. There is no fixed priority among the labels. When uncertain, establish scope and boundaries before comparing candidate types; request review if the type remains unresolved. [ESCO14] [ESCO-L] [ESCO-T]

SkillSpan's knowledge/skill examples can aid interpretation, but annotate Chinese sentences using this handbook's four definitions. [Z22] [Project rule]

Scope asks whether the source expresses an eligible language, knowledge, occupational-skill, transversal-competence, or education/qualification requirement. Application procedures such as submitting documents or taking a medical examination are outside scope in themselves. Boundaries identify the continuous source

substring; type identifies L, K, S, or T. Decide whether to annotate, then where the span begins and ends, and finally its type. [AP08] [Kr95] [Project rule]

Final annotation is flat and non-overlapping. The same source position cannot belong to two final entities, and each entity has one final type. Do not retain both a long entity and an entity inside it. Record alternatives separately for review. Once a long span is selected, for example, do not annotate one of its words again as a second final entity. [TKS02] [Project rule]

Procedure

Compare confusable types directly: K versus S contrasts knowledge content with occupational application; S versus T contrasts a specific occupational function with a general competence. [Project rule]

For boundaries, try removing the proposed words and ask whether the remaining text still expresses the same requirement. In a clear use context, a tool name can express the requirement to use that tool; generic “使用” (use) can remain outside when it adds no specific task. For activities such as “修电脑” (repair computers) and “写 SQL” (write SQL), removing the action changes an activity into a device or language name, so retain the action. Coordinated items may be separated when adjacent source text clearly supplies their shared qualifier and each item still expresses the requirement independently. Retain necessary shared wording if splitting leaves only an attribute, domain, or unclear generic term. When no unique boundary is justified, retain the alternatives and request review instead of choosing a reference answer arbitrarily. [Project rule]

If the passage is clearly in scope and its boundaries are settled, but its type remains uncertain, record the candidate types and reasons and request review. This handbook permits a provisional S only in this situation, together with adjudication_required. For example, if K and S remain plausible, retain both candidates and their rationales for the reviewer to resolve. The provisional S is not a confirmed occupational-skill label. Exclude the entire record from training until review is complete. If it is still unclear whether to annotate or which substring to select, settle those questions first; a provisional S cannot replace that decision. [Project rule]

Education and qualification names annotated as K

Related examples: [EX006](#), [EX007](#), [EX008](#) (Appendix A)

Annotate education, degree, and non-language qualification requirements as K. For example, “本科及以上学历” (bachelor's degree or higher), “大学本科及以上” (university bachelor's degree or higher), and “Bachelor Degree” all receive K. Language examinations and certificates follow the L rules. [Project rule]

This treatment draws on the SkillSpan guidelines by Zhang et al. (2022). Item 8 in Appendix B.2 of that paper assigns KNOWLEDGE to “Bachelor Degree” and annotates the following subject names separately. Items 2 and 3 likewise treat licences and certificates as knowledge annotations and retain components such as certificate or licence in their names. [Z22]

Qualifications and knowledge are not identical concepts. A qualification formally recognizes attainment of specified learning outcomes, which may include knowledge and skills. Assigning education and non-language qualification requirements to K is therefore a project annotation rule informed by the above research, not a classification prescribed by ESCO. [ESCO-Q] [EQF] [Project rule]

Retain wording needed to express the minimum education level, including “及以上” and “含以上” (or above; including higher levels). Select the whole “本科及以上学历” (bachelor's degree or higher), not merely “本科” (bachelor's level), so the minimum-level requirement is preserved. Retaining these Chinese threshold expressions is an explicit project boundary rule. [Project rule]

Treat education and duration of employment separately. “5 年以上” (five years or more) only expresses duration and is not a standalone entity; still annotate eligible education requirements in the same sentence. The shared word “以上” (above/more than) does not justify excluding both. [Project rule]

Communication and reporting annotated as T

Related examples: [EX009–EX013](#) (Appendix A)

Assess general communication skills, communication management, and reporting to clients as T. Assess natural-language listening, speaking, reading, writing, or proficiency as L. The former example “英语沟通→T” (English communication → T) is not a context-free rule: English expression ability is L; general communication or coordination is T when English merely describes the work setting. Annotate separate, non-overlapping expressions of both requirements separately. If the emphasis of one phrase remains unclear, request review instead of applying a fixed L/T priority. [Project rule]

Teaching contrasts, written to illustrate the criteria rather than taken from the dataset: “应聘者须具备英语口语表达能力。” (Applicants must be able to express themselves orally in English.) Here “英语口语表达能力” (oral English expression ability) is L. “应聘者须具备与客户沟通和协调的能力。” (Applicants must be able to communicate and coordinate with clients.) Here “与客户沟通和协调的能力” (ability to communicate and coordinate with clients) is T. The content of the requirement determines the type, not the isolated words English or communication, or the industry.

ESCO places report facts in communication category T4.1 and identifies it as reusable across contexts. T4.1 explains the conceptual source; it is not an additional annotation label. [\[ESCO-T\]](#)

SkillSpan has no separate T label, so its SKILL class is not identical to this project's S. The project separates general communication and related dispositions as T and combines them with S only for two-class evaluation. [Z22] [Say18] [Project rule] Say18 is the reference key for Sayfullina et al., not an annotation category.

General “沟通” (communication) remains T when expressed as a duty, such as “负责……工作” (responsible for ... work). Specific technical operations are assessed as S. [\[ESCO-T\]](#) [Project rule]

Occupational technical actions remain S: 接口对接 (interface integration), 信息梳理 (organizing information, with a specific business object), and problem handling with its necessary object. Person-to-person liaison is not S merely because it uses “对接” (liaise/interface). Annotate “协调外部资源” (coordinate external resources) as one T span, retaining its necessary object. [\[Z22\]](#) [Project rule]

Occupational abilities and general competences

Related examples: [EX014–EX017](#) (Appendix A)

To distinguish S from T, determine whether the requirement concerns a specific occupational tool, method, or task. General abilities such as communication and learning are assessed as T. Apply the research criteria below to research-related expressions; neither 能力 (ability) nor 研究 (research) alone determines the type. [ESCO-T] [ONET] [Project rule]

1. Occupational dependence: does the requirement explicitly depend on a tool, method used to perform a specific occupational task, or occupational object? If so, assess S; examples include 编码能力 (coding skills), 产品设计能力 (product-design ability), and 数据分析能力 (data-analysis ability).
2. General competence: does the expression mainly concern general cognition, self-management, communication, or interpersonal ability? If so, assess T; examples include 沟通能力 (communication skills), 学

习能力 (learning ability), and 沟通管理 (communication management). Use across several industries alone does not establish T; specific occupational functions such as project management still follow R17.

3. Classify research ability by the requirement expressed. General inquiry, thinking, or research ability is T. Specific occupational research activities, use of research methods, or identifiable research tasks are assessed as S. “论文 / 实验 / 课题” (paper/experiment/research project) are possible evidence, not mandatory words; their presence alone also does not establish S. Requirements limited to theory or methodological knowledge are K. If knowledge, practice, and general ability cannot be distinguished, record alternatives and request review. [Project rule]

Technical domains in context

Related examples: [EX018](#)–[EX021](#) (Appendix A)

Classify the same expression according to the requirement and local context. [Nav09] [Z22] [Project rule]
Nav09 refers to Navigli's survey of word-sense disambiguation in the references.

Context	Evidence	Label
Discipline certificate or major name	An explicit list of majors, certifications, disciplines, or education requirements.	K: 网络安全专业 (network-security major); 安全 / 网络 (security/networks) within certification names.
Object of planning maintenance or implementation	The text explicitly requires planning, operating, managing, implementing, or integrating networks/ICT systems, rather than merely describing their theory.	S: in “规划网络及 ICT.....运维计划” (plan networks and ICT ... operation/maintenance plans), select “网络” (networks) and “ICT 运维计划” (ICT operation/maintenance plans) as separate S spans.
Generic domain modifier	网络行业 (network industry), 安全领域 (security field), or ICT 市场 (ICT market), without an independent competence or explicit industry-background/experience requirement.	Do not annotate a mere industry or market modifier. For explicit applicant industry-background/experience requirements, apply the background rules in the following section.

Do not annotate “保证安全 / 确保安全” (guarantee/ensure safety) as an ability when it only states a goal or outcome. Where a specific safety operation or qualification name is expressed, classify the actual content; the shared word “安全” (safety/security) does not require a shared label.

Key distinctions

Related examples: [EX022](#)–[EX032](#) (Appendix A)

Identify the assigned work rather than classifying from a verb alone. Assess S when an identifiable occupational activity and object, together with duty, method, or implementation context, require the

applicant to perform the work. Do not separately label a desired state, numerical result, or performance target as a skill. “负责” (be responsible for) or a technical object alone is insufficient. “优化” (optimize), “提高” (improve), and “推动” (advance) are neither always included nor always excluded. Examine activities and goals separately when both occur; request review when evidence is insufficient. [Project rule]

- What to do (an action or method) → may be S.
- Desired outcome (a result or target) → do not annotate.

Expression	Interpretation	Treatment
分析推广效果 <i>English gloss: Analyze the effectiveness of promotional activities.</i>	An activity or method.	S
制定有效的推广策略 <i>English gloss: Develop effective promotional strategies.</i>	An activity or method.	S
优化曝光与转化率 <i>English gloss: Optimize exposure and the conversion rate.</i>	This expresses an optimization activity. Retaining only “曝光” (exposure) and “转化率” (conversion rate) would remove the necessary action and change the meaning into two metric names.	Retain “优化曝光与转化率” (optimize exposure and the conversion rate) as a complete S span, including the action and its objects.
保证流量/用户增长 <i>English gloss: Ensure traffic/user growth.</i>	An outcome or key performance indicator (KPI).	Do not annotate.
确保渠道目标达成 <i>English gloss: Ensure that channel targets are met.</i>	A desired outcome.	Do not annotate.

Do not annotate “保证 / 确保” (guarantee/ensure) when these merely promise a result. Check the following text for independent competences. Assess “质量保证” (quality assurance) as a whole when it names an occupational activity; do not mechanically remove “保证”. See R17. [\[PB05\]](#) [Project rule]

Following content	Treatment	Example
A result or state noun only	Do not annotate the phrase.	增长、流量、目标、业绩、稳定、质量、满意度、达标、合规、可靠、高效、优质；保证系统稳定；确保服务质量；保证安全 <i>English gloss: Growth, traffic, targets, performance, stability, quality, satisfaction, meeting standards, compliance, reliability, efficiency, high quality; guarantee system stability; ensure service quality; guarantee safety.</i>
A desired outcome in this example	Extract no entity in this example.	“确保数据备份策略有效”只说明期望达到的结果，没有明确要求掌握某项知识或执行某项操作，本例不提取实体。不能据此排除所有含“确保”的句子；若句中另有明确的职业活动，仍按相应规则标注。 <i>English gloss: “Ensure the data-backup strategy is effective” states a desired outcome without an explicit knowledge or operational requirement. Extract no entity in this example. This does not exclude all sentences containing ensure; annotate any explicit occupational activity according to the relevant rule.</i>
Action plus outcome	Annotate the action only.	保证按 SLA 完成故障响应 → 故障响应 S <i>English gloss: “Ensure incident response is completed according to the SLA” → 故障响应 (incident response), S.</i>

Metrics following “分析 / 制定 / 优化” (analyze/develop/optimize) can be objects of work and may form S spans. A preceding “保证” (guarantee) does not make the entire phrase automatically empty or automatically S.

Boundaries for background and experience

Related examples: [EX033](#)–[EX037](#) (Appendix A)

This section replaces the blanket advice to usually remove “经验” (experience) in v4.2.9. Retain or exclude it according to whether removal preserves the same requirement. Duration and preference expressions such as “优先” (preferred) are not entities in themselves. [Project rule]

Condition	Treatment	Example
Generic experience without a specific domain or activity	Do not annotate.	工作经验、项目经验、丰富经验 <i>English gloss: Work experience; project experience; extensive experience.</i>
Explicit industry-background or industry-work-experience requirement	Annotate each core domain name as K; exclude generic experience/background wording, duration, and preference conditions.	有银行、咨询、服务行业工作经验优先考虑 → 银行 K、咨询 K、服务行业 K <i>English gloss: “Experience in banking, consulting, or the service industry is preferred” → 银行 (banking), K; 咨询 (consulting), K; 服务行业 (service industry), K.</i>
A specific occupational activity that remains complete without experience wording	Assess the activity as S. Exclude “经验” (experience) when the remaining text still expresses the same activity; retain necessary experience wording when removal would change the requirement.	大项目售前、数据分析、大数据开发、大数据项目实施、BI 项目实施 <i>English gloss: Presales work on large projects; data analysis; big-data development; big-data project implementation; BI project implementation.</i>
Experience with a function or product category where removing experience leaves only a category and changes the meaning	Retain experience and annotate the complete phrase as S.	产品经验、内容型产品经验 <i>English gloss: Product experience; experience with content-based products.</i>

When the applicant is explicitly required to have industry background, assigning the core domain name to broad K is a project rule for representing background knowledge. First confirm that an applicant-background requirement is actually expressed. Company activities, the job's industry, product markets, personal interests, and user identities do not by themselves establish such a requirement. Retain “行业” (industry) in “服务行业” (service industry) to make the domain meaning clear.

When to retain the colloquial word 会

Related examples: [EX038](#)–[EX040](#) (Appendix A)

When “会” (know how to) is integral to a colloquial interpersonal competence, retain it and annotate the complete phrase as T. [Project rule]

When “会” only introduces the ability to perform the following tool-based task or duty, exclude it. SkillSpan annotates attitudes as skills and excludes function words such as articles. [Z22]

Source text	Treatment
会聊天、会说话、会来事、会做人 <i>English gloss: Can hold a conversation; know how to speak appropriately; be socially adept; know how to get along with people.</i>	Annotate the complete phrase as T, including “会”.
会使用 Excel、会修电脑、会写 SQL <i>English gloss: Able to use Excel; know how to repair computers; know how to write SQL.</i>	Confirmed annotation: Excel S; 修电脑 (repair computers) S; 写 SQL (write SQL) S. In the first expression, “使用” only signals generic tool use; the actions in the other two distinguish specific work. 会 and 能够 express capability, while 熟练 expresses proficiency; normally leave them outside the entity. 使用 (use) is an action, not a proficiency modifier. Generic tool use may be represented by the tool name: both 会使用 Excel and 熟练使用 Excel yield Excel, S. Retain necessary actions and objects for a specific task: 熟练使用 Excel 制作财务报表 yields 使用 Excel 制作财务报表 (use Excel to prepare financial reports), S; 能够配置数据库 yields 配置数据库 (configure databases), S. 熟练掌握数据库原理 yields 数据库原理 (database principles), K. Proficiency does not determine the type. Apply the separate rules for language levels and T expressions of willingness.

For “善于 / 熟悉 / 掌握 / 了解 / 精通 / 具备 / 有” (be good at/be familiar with/have command of/understand/be proficient in/possess/have), select the following content rather than the introductory expression: 善于与人沟通 → 与人沟通 (communicate with others); 有服务意识 → 服务意识 (service-oriented mindset). [Z22] Appendix B.3.6 treats advanced knowledge of and proficient in as triggers outside the span.

Selecting source-text spans

Related examples: [EX041](#)–[EX049](#) (Appendix A)

1. Select continuous, complete source substrings without correcting spelling or adding words. “支持服” and “操作系统的问” are incorrect because they truncate words (fragments of support service and operating-system problem). The technical check is sentence[start:end] == span: slicing the recorded positions must reproduce the entity text exactly. [\[Z22\]](#) [\[D18\]](#) [\[AP08\]](#) [Project rule]

2. Select the shortest complete span that preserves meaning. Semantic completeness determines length. Python, 计算机专业 (computer science major), and 沟通能力 (communication skills) can each stand alone. Annotate independent duties separately, retaining each complete meaning. [\[Z22\]](#) [\[AP08\]](#) [\[Kr95\]](#) [Project rule]
3. Separate items joined by “、”, “和”, “与”, or “或” (enumeration comma/and/and/or) only when each independently expresses the original requirement. Retain a complete span if splitting truncates words, removes necessary shared wording, or changes meaning. For example, splitting “优化曝光与转化率” (optimize exposure and the conversion rate) into two metric names removes the optimization activity. [\[Z22\]](#) [\[UD20\]](#) [Project rule]
4. Exclude introductory “熟悉 / 掌握 / 了解 / 精通 / 具备” (be familiar with/have command of/understand/be proficient in/possess), and select their content. [\[Z22\]](#) [Project rule] Colloquial interpersonal expressions such as “会聊天” (can hold a conversation) are exceptions; see the preceding section. [Project rule]
5. Classify programming languages, frameworks, software, and tools by context. A tool name in occupational use may be S. Explicit course content, principles, fundamentals, or syntax are K as complete knowledge phrases, such as “Python 语言原理” (principles of the Python language). If scope and boundaries are clear but type is unresolved, record a provisional candidate and request review. [\[EQF\]](#) [\[ESCO14\]](#) [\[Z22\]](#) [Project rule]
6. Natural-language names, proficiency, and examinations, such as 英语六级 (College English Test Band 6), CET-6, and 日语 N2 (Japanese proficiency at N2 level), are L. Non-language technical or occupational qualifications such as ISO and OCJP are K under project rules. Older versions used different labels; do not overwrite archived data on that basis. [\[ESCO-L\]](#) [\[ESCO-Q\]](#) [Project rule]
7. Applicant-facing procedures such as application, medical examination, public announcement, and eligibility checks are not annotated, nor are benefits, statutory insurance/housing-fund benefits, shifts, locations, or slogans in themselves. Record spans: [] only after confirming that the whole sentence has no target entity. Insufficient evidence, missing context, and disagreement are unresolved states; see R19. Recruitment, interviewing, and background checks performed by recruitment staff may be occupational activities, so do not exclude them by keyword alone. [\[Z22\]](#) [Project rule]
8. Exclude appearance, physical health, numerical durations (retain the ability itself), and driving licences for non-driver roles. [\[Z22\]](#) [Project rule]
9. Final annotation is flat and non-overlapping: one type per span, and at most one entity per source character. [\[TKS02\]](#) [\[FM09\]](#) [\[Yu20\]](#) [\[AP08\]](#) [Project rule]
10. The same name can receive different types in different contexts. Assess SQL/Python used in actual work as tool-name S. When the requirement concerns courses, principles, fundamentals, or syntax, annotate the complete knowledge phrase as K rather than the tool name alone. See the following section. [\[ESCO14\]](#) [Project rule]
11. Human-confirmed source-character positions are authoritative. jieba is a Chinese word-segmentation tool; its token boundaries can support checking or an alternative display/evaluation representation, but must not automatically replace human span boundaries. Record competing types and nested or crossing span candidates separately for review. The final annotation retains only confirmed, non-overlapping entities. [\[AP08\]](#) [\[D18\]](#) [\[FM09\]](#) [\[Z22\]](#) [Project rule]
12. An activity may be S; a desired outcome, goal, or KPI is not annotated. See the key distinctions above. [\[Z22\]](#) [Project rule]

Tool names as S and knowledge expressions as K

Related examples: [EX050–EX057](#) (Appendix A)

Look for source evidence when classifying a tool name. Explicit use, installation, or development, hands-on experience, or a tool list linked to an adjacent occupational task supports S. Courses, principles, fundamentals, or syntax support K as a complete knowledge phrase. If the text only says familiar with/have command of plus a name, without such evidence, record candidate types and reasons and request review. Do not decide solely from the job title or default all tools to S or K. [Project rule] [EQF]

The following table contrasts types and boundaries. Some rows are only course or textbook fragments. First establish that they express applicant requirements before extracting entities; the occurrence of a knowledge term such as Python alone does not prove a competence requirement in the whole advertisement. Full cases appear in Appendix A, Part II.

Typed F1 evaluates both span selection and classification. Under typed exact matching, both source start/end positions and label must match the reference. Selecting the correct text but assigning S instead of K is not a correct match. For relaxed matching, specify the permitted boundary differences. Before comparing datasets, check whether label sets, boundary rules, and matching criteria agree. [Z22] [Project rule]

Sentence	Annotation	Avoid
掌握常用算法编程语言，如 R, Python, C 等 <i>English gloss: Have a command of programming languages such as R, Python, and C that are commonly used to implement algorithms.</i>	R S; Python S; C S. Here “常用算法编程语言” (programming languages commonly used to implement algorithms) only introduces the specific list and is not separately annotated.	Changing items after “如” (such as) to K; adding nested K spans inside a long S span.
熟悉使用 Word, Excel, PPT <i>English gloss: Be familiar with using Word, Excel, and PowerPoint.</i>	Word S; Excel S; PPT S	Annotating “熟悉使用” (be familiar with using).
计算机专业课程包括 Python、数据结构 <i>English gloss: Computer science courses include Python and data structures.</i>	Type illustration, not a complete extraction answer: Python and 数据结构 (data structures) may be K in a course-knowledge context. The fragment alone does not establish “计算机专业” (computer science major) as an education or subject requirement.	Assigning S to Python.
了解 Python 语言原理 / 解释器实现 <i>English gloss: Understand Python language principles / interpreter implementation.</i>	Python 语言原理 (Python language principles), K; 解释器实现 (interpreter implementation), K.	Selecting only Python and omitting the knowledge content “语言原理” (language principles).

Sentence	Annotation	Avoid
具备 Python 基础知识（后文不再要求开发） <i>English gloss: Have basic knowledge of Python (with no subsequent development requirement).</i>	Python 基础知识 (basic knowledge of Python), K.	Selecting only Python; including “具备” (possess).
教材：Python 语法与标准库 <i>English gloss: Textbook: Python syntax and the standard library.</i>	In an applicant knowledge requirement, annotate “Python 语法与标准库” (Python syntax and the standard library) as one K span. Retain the shared Python qualifier so that the standard library remains identifiable. A textbook title alone does not establish an applicant requirement.	Selecting only Python.
了解 SQL 原理 / 用 SQL 查库 <i>English gloss: Understand the principles of SQL / use SQL to query a database.</i>	Annotate “SQL 原理” (principles of SQL) as K and “SQL 查库” (querying a database with SQL) as S. Retain the specific query action and exclude 用 (using). Preserve the source space.	Treating the two contexts as one fixed type, or creating an entity absent from the source.

Additional guidance on scope and complete expressions

Related examples: [EX058](#)–[EX064](#) (Appendix A)

Provenance: this section arose from discussion during correction of approximately 100 records. That does not establish 100 verified, distinct final human reference records. Keep confirmed rules and pending records separate. These wording changes do not alter historical annotations or experimental results. Maintenance records are in Appendix B.

1. Establish scope before type. Check whether the expression concerns an applicant's competence, knowledge, or qualification before selecting boundaries and classifying. Big data or cloud computing listed as company technical advantages are not therefore K. Do not extract 网络 (networks) from 网络课程 (online courses) mentioned as a reimbursement benefit. Company encouragement of self-drive is not automatically an applicant requirement. When duties and benefits coexist, assess each clause rather than clearing the whole sentence.
2. Methods are not always K. Technical methods used to perform occupational tasks are S; theories, principles, and standards are K. Examples: Linux 内核调试方法 (Linux kernel debugging methods), S; Linux 系统原理 (Linux system principles), K; 编码规范 (coding standards), K. Exclude 熟悉 (be familiar with) and 常见的 (common), but do not remove 方法 (methods) merely to shorten the span if this changes the required object.
3. Completeness takes priority over brevity. Do not shorten 协调外部资源 (coordinate external resources), T, to 协调 (coordinate), or remove necessary qualifiers from 严谨的工作态度 (a careful and meticulous

approach to work), T. When product documentation is the object of a writing activity, do not add nested K. System stability is a system property, not a person's T.

4. Distinguish independently meaningful items joined by and/or from items that need shared wording. Annotate independently identifiable tools separately. Keep one span when shared wording is necessary, as in 口头、书面表达能力 (oral and written expression ability), T, and KVM 的开发和实施 (KVM development and implementation), S. Do not split mechanically at and, or combine independent duties into one S. Complex lists with omitted shared words still require review; see the issue register.

5. Select source text only: do not add words or truncate them. If the source only contains Memory, do not create Memory 开发经验 (Memory development experience). Do not cut 缓存 (cache) or RocketMQ mid-word. Preserve spelling, capitalization, and spaces. A normalized name may be recorded separately but cannot replace the selected source text.

6. Category nouns do not automatically add K. Check whether category expressions such as frameworks, databases, and design tools independently express a knowledge requirement. Being a noun is insufficient. Distinguish practice from principles. Register uncertain category/tool boundaries instead of claiming that a uniform answer has already been established.

7. Candidate annotations require review. Model suggestions and comparison records are not automatically human reference labels. Retain the source, candidate spans, confirmed spans and types, rationale, rule IDs, reviewer, and version. Unresolved records remain pending. Corrections made after viewing model suggestions are not original independent blind annotations for agreement calculation. [Project rule]

These are the manual annotation rules and their explanations. Pending entries in the register continue through the difficult-case review procedure.

Contrasting examples

Related examples: [EX065](#)–[EX076](#) (Appendix A)

Sentence	Handbook B	Avoid	Source
熟悉使用 Word, Excel, PPT 等办公软件 <i>English gloss: Familiar with using Word, Excel, PowerPoint, and other office software.</i>	Word S, Excel S, PPT S	Including 熟悉使用 (be familiar with using), or selecting one long S span.	[Z22] [Project rule]
本科及以上学历, 大学英语 6 级 <i>English gloss: Bachelor's degree or higher; College English Test Band 6.</i>	本科及以上学历 (bachelor's degree or higher), K; 大学英语 6 级 (College English Test Band 6), L.	Omitting the education requirement, or assigning K to the English examination.	[Z22] [ESCO-Q] [ESCO-L]
维护和支持服务 <i>English gloss: Maintenance and support services.</i>	维护 (maintenance), S, plus 支持服务 (support services), S, with complete word boundaries.	Truncating 支持服务 (support services) to 支持服.	[Z22] [Project rule]

Sentence	Handbook B	Avoid	Source
五险一金，带薪年假 <i>English gloss: Five types of social insurance and a housing fund; paid annual leave.</i>	Empty annotation: [] (confirmed to contain no target entity).	Forcing benefits into the annotation.	[Z22]
掌握……如 R, Python, C <i>English gloss: Have command of ... such as R, Python, and C.</i>	R, Python, and C each receive S in an occupational-use context.	Assigning K to all items, or selecting only the long category expression.	[Project rule]
了解 Python 语言原理 <i>English gloss: Understand Python language principles.</i>	Python 语言原理 (Python language principles), K.	Selecting only Python.	[ESCO14] [Project rule]
分析推广效果，保证流量/用户增长 <i>English gloss: Analyze the effectiveness of promotional activities and ensure traffic/user growth.</i>	分析推广效果 (analyze the effectiveness of promotional activities), S; exclude 保证流量 / 用户增长 (ensure traffic/user growth) as an outcome.	Annotating the pure outcome 保证流量 / 用户增长 (ensure traffic/user growth) as S.	[Z22] [Project rule]
编码能力 / 沟通能力 / 科研能力 <i>English gloss: Coding skills / communication skills / research ability.</i>	编码能力 (coding skills), S; 沟通能力 (communication skills), T; 科研能力 (research ability), T, only when it means general research ability. Assess specific research activities as S and methodological knowledge as K using the main rules.	Treating every ability to do something as S.	[ESCO-T] [ONET] [Project rule]
认证清单「安全」；规划网络……运维计划 <i>English gloss: Security in a certification list; planning networks ... operation/maintenance plans.</i>	Distinguish by context: knowledge/qualification s in a certification list are K; explicit network-planning or maintenance activities are S. This row illustrates context only; determine actual boundaries from the full source sentence.	Assigning one fixed type to each word.	[Nav09] [Project rule]

Sentence	Handbook B	Avoid	Source
大项目售前经验 <i>English gloss: Experience in presales work on large projects.</i>	大项目售前 (presales work on large projects), S; exclude 经验 (experience).	Retaining experience in the whole phrase unnecessarily.	[Project rule]
负责与客户的沟通管理工作, ……客户汇报工作 <i>English gloss: Responsible for managing communication with clients, ... and reporting to clients.</i>	沟通管理 (communication management), T; 客户汇报 (reporting to clients), T.	Changing the label to S because of 负责 (responsible for).	[ESCO-T] [Z22]
Scope and boundaries are established, but comparing candidate types has not resolved the type.	Record provisional S, the candidate types, and pending-review status; do not use it directly as a final training label.	Replacing contextual judgement with a label ordering.	[Z22] [AP08]

Supplementary rules R08 to R13

R08–R13 add guidance on experience, context, unresolved records, and protection of source text. Historical prompts and execution batches are documented in Appendix B; wording changes here do not alter those records.

R08 Experience suffixes and source-text limits

Related examples: [EX077](#)–[EX078](#) (Appendix A)

If the source explicitly requires specific practical experience and removing 经验 (experience) leaves only a category, domain, or competition name and changes the meaning, retain the complete expression: 工艺经验 (process-engineering experience) and 机器人竞赛经验 (experience in robotics competitions) → S. When a specific activity such as 数据分析 (data analysis) remains complete without experience, exclude the suffix. Apply the existing broad-K rule to industry background; exclude generic experience.

Retaining words does not mean adding them. Do not add 经验 (experience) to a span unless it is part of the selected continuous source text. When coordinated items share a following experience phrase, first apply R14 and R23 to test whether coordinated items can be separated without meaning loss in local context. Retain the necessary shared span only if splitting changes the meaning; register unresolved boundaries. Interpret experience together with its activity or requirement.

ESCO's skill definition (<https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/skill>) emphasizes applying knowledge and know-how to complete tasks. Competence (<https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/competence>) concerns demonstrated use of knowledge, skills, and personal abilities in work or learning contexts. These definitions do not prescribe the type or boundary of the Chinese suffix 经验 (experience). Retaining that suffix and assigning S here are project annotation decisions, not labels assigned to these Chinese expressions by ESCO. [\[ESCO-C\]](#)

R09 General problem solving and practical methods

Related examples: [EX079](#)–[EX080](#) (Appendix A)

解决问题 (solve problems) and 分析问题 (analyze problems), without a specific occupational object, are T. For example, 善于解决问题和分析问题 (be good at solving and analyzing problems) yields two T spans, excluding 善于. Reassess S when a specific technical or business object is supplied. A technical job advertisement does not turn all general competences into S.

嵌入式软件设计方法 (embedded-software design methods) is S in this example because it names a method used to perform a specific occupational task in context; 计算机原理 (computer principles) in the same sentence is K. Keep introductory familiar with, understand, and proficient outside the spans; classify the required content.

R10 Additional cases and review status

Related examples: [EX081](#)–[EX085](#) (Appendix A)

The first four treatments below are confirmed. The fifth requires context and must not always be treated as containing no entities.

1. 将公司技术能力/产品能力转化为商业语言 (translate the company's technical/product capabilities into business language) → one S. Technical capabilities and product capabilities are objects of the translation activity, not additional nested K entities.
2. 提高广告系统的用户体验、投放能力及变现效率 (improve the advertising system's user experience, delivery capability, and monetization efficiency) is excluded when it only states desired system outcomes. If the full requirement separately specifies improvement work, reassess it under the activity rules. 对解决挑战性问题的热情 (be enthusiastic about solving challenging problems) is one T, without nested problem-solving S. In 数据结构和算法基础扎实 (have a solid foundation in data structures and algorithms), annotate 数据结构 (data structures), K, and 算法基础 (algorithm fundamentals), K, excluding 扎实 (solid).
3. 嵌入式软件设计方法 (embedded-software design methods) → S under R09; 熟练 (proficient) is not the deciding condition.
4. 构建新产品/新工艺转生产阶段转产流程 (establish the production-transfer process for new products/processes entering production) → one S. 承接公司新产品/新工艺的转产工作 (undertake production transfer for the company's new products/processes) → 新产品/新工艺的转产 (production transfer of new products/processes), S. Exclude 提升工艺团队转产能力 (improve the process team's production-transfer capability) if it is only a goal without an implementation method.
5. 能力提升: 媒体平台特性/互联网思维/运营推广方法论/接触前沿动态 (capability development: media-platform characteristics/internet thinking/operations and promotion methodology/exposure to current developments) requires context. Exclude confirmed benefits or post-employment prospects; assess each item if they are applicant requirements; otherwise register unresolved scope. A capability-development heading alone does not justify an empty annotation.

R11 Unresolved scope boundaries and types

Related examples: [EX086](#)–[EX088](#) (Appendix A)

- Unresolved scope: retain established spans separately as review evidence and record disputed content and missing context. While any scope issue remains unresolved, do not mark the whole record as confirmed empty [], or send it into training merely because some entities are known. See R19.

- Unresolved boundaries: record each continuous, complete candidate span and its rationale, and request review. A preferred candidate may be noted, but remains a candidate until reviewed; it is not a final answer or complete training annotation.
- Unresolved type: after scope and boundaries are established, compare types in pairs. Use provisional S only if uncertainty remains, retaining alternative types and reasons.
- Place unresolved records on the review list and exclude the entire record from training for now. Authorized model review may be used for Silver labels. Continue to isolate unresolved records; see R19–R20. Save changes as a separate version after quality checks.

R12 Model identity and execution records

Record the model identifier supplied by the system or a reliable environment, with its source; use unknown when unavailable. Keep a model's self-reported name separately. Also retain the prompt version and checksum, input/output checksums, execution time, and batch ID. Historical gpt6_* field names are compatibility identifiers only.

R13 Character positions and source protection

Character intervals [start,end) are zero-based and include the start position but exclude the end position. Chinese characters, Latin letters, and spaces all occupy positions. Check by Unicode code point that text[start:end] exactly matches the selected source text. In 用 SQL 查库 (use SQL to query a database), the letter S occurs at position 2; the confirmed S span SQL 查库 has interval [2,8). JSONL stores one record per line. These offsets are not UTF-8 byte offsets and need not match a software interface's UTF-16 counts. Preserve leading spaces, spelling, and case. Sort final spans by start position, with no duplicates or overlaps. Verify the export tool's counting convention separately; see D14 in Appendix C.

Check structure, character positions, and semantic decisions separately. Save label changes as a separate version and retain original records. Historical model-review interfaces and permitted update fields are listed in Appendix B.

R14 Meaning-preserving splits and necessary long spans

Related examples: [EX089](#)–[EX092](#) (Appendix A)

R14 is a supplementary rule first recorded on 2026-09-07. This edition retains the rule and subsequently confirmed type treatments; historical prompts and execution records retain their original versions.

Before splitting coordinated items, ask what would be removed, where the sentence supplies that information, and whether the remaining words still express the same requirement. The table below explains the included examples. Tool names and occupational activities can be separate entities when supported by the immediate sentence context. If an attribute or domain name needs a shared action or head word to express the requirement, retain that component. Do not insert absent words. Determine the type of a retained long span separately. If the answer remains uncertain, request review under R11. [Project rule]

Boundary comparisons

Example and treatment	Information that can remain in context	Reason for retaining or omitting words
会使用 Excel → Excel S <i>English gloss: Able to use Excel → Excel, S.</i>	会使用 (know how to use) establishes general tool use.	Excel identifies the tool; 使用 (use) adds no separate specific task. 修电脑 (repair computers) and 写 SQL (write SQL), in contrast, retain their repair/writing actions.
Case 120 → annotate 架构设计 (architecture design) and 运维 (operations and maintenance) separately as S.	The preceding item specifies the cloud-computing infrastructure service platform.	运维 (operations and maintenance) is an identifiable occupational activity; the preceding item in the same sentence supplies its object. Do not insert the platform name.
Case 117 → 分析 (analysis), S.	The immediately preceding text specifies difficult live-network problems involving cloud products.	The technical object of analysis is explicit. Do not apply this decision to an isolated 分析 (analysis) without object evidence.
Case 129 → retain the long specialist-testing-techniques span as S.	The list items are not independent skill answers.	Selecting only 安全 (security) or 性能 (performance) leaves attributes or domains; the shared 测试技术 (testing techniques) makes them occupational requirements.
Case 103 → 产品的探索 (product exploration), S.	The fuller product context remains in the source sentence.	Retain the product object to distinguish occupational product exploration from general exploration. The confirmed span is 产品的探索, not 探索 alone.
Python 语法与标准库 (Python syntax and the standard library) → one K span in a knowledge requirement.	Python qualifies both syntax and the standard library.	Retain the shared qualifier to make the library identifiable. A textbook title alone is not an extraction case.
Case 194 → BI, S.	The items share the following 相关工作经验 (relevant work experience).	BI is independently identifiable within this practical-experience requirement. The longer continuous phrase does exist; it is simply not selected here.

Confirmed boundaries and types

案例 130 原句：5.熟悉 Linux 基本操作和原理，在云计算、深度学习等领域者有工作经验优先；

English gloss: Case 130: 5. Be familiar with basic Linux operations and principles; practical work experience in areas such as cloud computing and deep learning is preferred.

Annotate Linux 基本操作 (basic Linux operations), S; 原理 (principles), K; 云计算 (cloud computing), S; and 深度学习 (deep learning), S. The last two types follow the local practical-work-experience requirement, which remains in context. The current answer uses four independent spans.

Confirmed boundaries and types

案例 134 原句：任职要求： 1、熟练掌握自然语言处理 NLP 相关理论与技术方法；

English gloss: Case 134: Job requirements: 1. Have a strong command of theories and technical methods related to natural language processing (NLP).

The confirmed answer separates 自然语言处理 NLP 相关理论 (theories related to natural language processing/NLP), K, from 技术方法 (technical methods), K. The two items are joined by 与 (and), and NLP also qualifies 技术方法. The sentence describes theoretical and methodological knowledge. Do not create NLP 技术方法, which does not occur as a continuous substring. The earlier single K span was replaced by two K spans.

Diagnostic records for necessary long spans

When explaining a long span, record the shared action or object, the information splitting would lose, and the reason for retaining it. Cases used to develop rules or correct model output serve teaching or diagnostic purposes. Independent testing requires samples not involved in those processes.

Historical revision counts, confirmation methods, and separately saved versions are listed in Appendix B.

R15 Knowledge methods tools and distinct requirements within a sentence

Related examples: [EX093](#)–[EX097](#) (Appendix A)

1. Knowledge content is K; application or operation is S. Classify the object and requirement in the current clause, not the triggers familiar with, understand, or have command of. A technical job context does not make everything S.
2. Theory and methods. In case 128, 软件工程理论和方法 (software-engineering theory and methods) is a general knowledge system and one K span. The later clause explicitly requires testing practice: 软件测试的技术 (software-testing techniques), S;
理论 (theory), K; 方法 (methods), S; 白盒测试 (white-box testing), S; 黑盒测试 (black-box testing), S. Case 134 lacks the same practice evidence, so 技术方法 (technical methods) is K. Linux 内核调试方法 (Linux kernel debugging methods) and 嵌入式软件设计方法 (embedded-software design methods) directly name methods used to perform specific occupational tasks and are S. This is consistent with the distinction from general methodological knowledge.
3. Tools involving both knowledge and practice. Case 185 requires understanding principles and having hands-on experience. Its historical labels for HDFS, MongoDB, Redis, Memcached, Kafka, and Dubbo are S; the preceding six knowledge items remain K. Alternative K interpretations of the tools are retained in review records, separate from the established six preceding K answers. The practice requirement applies locally to the tool group. Preserve original spelling.

4. Technical categories require case-specific interpretation. In [case 102](#), jvm, 关系型数据库 (relational databases), Nosql, RPC 框架 (RPC frameworks), 消息队列框架 (message-queue frameworks), and Zookeeper are S in the occupational tool-stack context; 流计算 (stream computing) is K as general technical knowledge. This is the project's conservative interpretation of this case.

5. A qualifier may apply to nearby items joined by and/or, but not automatically to the whole advertisement. In case 130, practical work experience qualifies both cloud computing and deep learning. The shared experience wording may remain outside the entities while still guiding their interpretation. Explicit industry background follows the project rule for broad K; generic experience is excluded. Remove experience from complete activity names, but retain it when needed to preserve meaning under R08. Apply these conditions separately.

6. Complete knowledge phrases and local ellipsis. Retain explicit phrases such as 算法基础 (algorithm fundamentals). In case 130, 原理 (principles) refers locally back to Linux. Do not create discontinuous Linux 原理 or combine operations and principles merely to force one type.

These are rules confirmed for Chinese annotation in this project, with conceptual reference to ESCO Knowledge (<https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/knowledge>) and Skill (<https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/skill>). [\[ESCO14\]](#)

R16 Willingness dispositions and research ability

Related examples: [EX098](#)–[EX101](#) (Appendix A)

- Explicit willingness to learn or develop may be T. In case 160, 有意愿在 nlp 相关领域深入研究 (be willing to pursue in-depth research in NLP-related areas) is one T. Retain the willingness meaning; do not shorten it to 深入研究 (in-depth research), S, or nest NLP knowledge inside it. This is an exception for the complete meaning of willingness plus action, not a reason to include 有 (have) in ordinary 有服务意识 (have a service-oriented mindset).
- Explicit job-related learning/development motivation differs from mere product preference. Do not add an entity for 喜欢高科技电子产品 (like high-tech electronic products) in case 169. This does not settle historical search-engine enthusiast or advanced-user identity cases automatically.
- In case 195, 对商业和业务逻辑敏感 (sensitivity to commercial and business logic), 数据思维 (data-oriented thinking), and 数据决策意识 (awareness of data-driven decision-making) are T, as is 数据敏感度 (a feel for data) in case 197's disposition list. A domain word does not automatically turn general thinking or awareness into S. Specific data-analysis or modelling skills are assessed separately as S.
- Case 158 retains the historical T for 学术研究 (academic research), which used the earlier research criteria. Do not reaffirm it merely because paper, experiment, or research project is absent. Before using the case under revised rules, check whether the research-experience wording supports a specific activity and record the confirmation. Do not infer a particular competition from unspecified 竞赛经验 (competition experience).

For the concept of willingness to learn, see §5.3 of the ESCO expert-group report (<https://esco.ec.europa.eu/system/files/2022-05/MSWG%2014-04%20Report%20of%20the%20expert%20group%20on%20transversal%20skills%20and%20competences.pdf>). Retaining Chinese willingness wording is a project span decision, not a verbatim rule from that report. [ESCO-TG]

R17 Occupational functions and general coordination

Related examples: [EX102–EX107](#) (Appendix A)

- Use across industries does not imply T. Project management in case 152 and product-development/translation-project contexts names an occupational function, S. General organizational management ability and leading a team are T. Project-management courses or principles may be K. ESCO's reusability levels (<https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/skill-reusability-level>) provide conceptual distinctions, not direct prescriptions for these Chinese labels. [\[ESCO-R\]](#)
- Classify resources and interfaces by the actual activity. In case 153, 翻译项目资源接口 (resource liaison for translation projects) denotes a specific resource-coordination duty within translation projects and is S. In case 154, 开发储备适用资源 (develop and build a pool of suitable resources) is also a specific occupational activity, S. 协调外部资源 (coordinate external resources) is T when it expresses general coordination. A domain name alone does not establish S.
- Case 124's original table separately labels 深度学习平台的测试 (testing of a deep-learning platform), S, and 质量保证 (quality assurance), S. These name testing and quality-assurance work, not a mere outcome promise. Do not mechanically remove 保证 (assurance) from the occupational name. See case 124 for the complete source.
- For words such as 分析 (analyze), establish what is being analyzed from the current sentence. In case 117, 分析 follows 云产品现网疑难问题 (difficult issues in live cloud-product networks), so it expresses technical work and is S. In case 127, reviewing problems and proposing improvements express general reflection on work and are T. 质量意识 (quality awareness) and 测试理念 (a testing mindset) are T in that sentence because they express work-related awareness and attitudes; developing project test plans and testing strategies is S. Assess each coordinated item by its meaning. Coordination alone does not justify assigning the same type.
- Retain necessary objects. The three long technical activities in case 102 and 技术文档的编制及审核 (preparing and reviewing technical documentation) in case 179 follow the complete-case tables. Do not automatically extract the preceding requirements-analysis, high-level-design, and detailed-design stages as separate duties. In case 103, the confirmed S span is 产品的探索 (product exploration), not 探索 (exploration) alone; see D02 in Appendix C. Case 120 lists architecture design and operations separately, as shown in its table.
- Retain shared key words. The specialized testing techniques in case 129, consultancy across project stages in case 143, and design lists in case 167 retain their shared head and are labeled S. Do not detach their objects and mistake them for separate knowledge requirements. Do not insert action words absent from the source.

R18 Review records and data versions

Related examples: [EX108](#) (Appendix A)

1. Distinguish human review, conditional confirmation, and model review; do not describe all of them as independent blind annotation. Historical batch methods and counts are in Appendix B.
2. Change records should identify revised labels or boundaries, unchanged records, and any unresolved cases. Counts of revisions are not measures of accuracy or agreement. Historical counts appear in Appendix B.
3. Preserve the original execution procedures, model information, labels, and version records. Store reasons for changes and file checksums with each revised copy.

4. Retain candidate answers, review rationales, and current status for difficult records. Handle new questions under R11; see Appendix B for the completion status of historical batches.
5. Cases used to develop rules or train models may support related diagnostic analysis. Independent testing requires separate safeguards against reuse. Trace later experiments through their own run records.
6. Record versions and release status separately for the Word reading copy, English edition, GitHub, Overleaf, and machine prompts. Editing this handbook does not mean those materials have been synchronized.

R19 Separate empty and difficult records

Related examples: [EX109](#)–[EX112](#) (Appendix A)

This rule standardizes record states for human annotation and model review; it adds no S/K/T/L entity type. Empty is a record state, not a fifth entity class. First determine whether an extractable span exists in the current source, then its boundaries and type.

Use `confirmed_empty` with final labels [] only after reading the source and confirming that it contains no target entity. A sentence solely describing benefits, application procedures, or company promotion may qualify after checking. An empty model output still requires source inspection: it may reflect genuinely absent entities, missed entities, or a failed response. If the source explicitly includes 本科及以上学历 (bachelor's degree or higher), annotate that K requirement even when the model outputs []; do not confirm an empty record.

Use `difficult_or_undetermined` for uncertain scope, boundaries, or types, missing characters, garbled text, or missing essential context. In review records, final labels is null and retained candidates are stored separately in `candidate_labels_preserved`. If legacy compatibility requires [] or partial candidates, also preserve the unresolved status and isolate the whole record. Do not treat an undecided answer as a confirmed negative example, or allow a trainer to convert null automatically into [].

Confirmed sentence-level decisions: record 1928-s0005, 功能解决对应问题; (the functionality addresses the corresponding problems), states only a generic effect and is empty. Records 1996-s0012 and 1996-s0013, 2.其它职责同上。 (2. Other duties as above), contain only a cross-reference, with no skill entity stated in the sentence, and are also empty. These decisions concern extraction of continuous spans within a sentence; they do not imply that no duties appear earlier. If an explicit skill appears alongside 同上 (as above), extract it rather than clearing the entire sentence.

Conversely, 基于分布式及大数据平台的优先考虑。 (those based on distributed and big-data platforms are preferred) contains possible technical mentions but lacks the essential activity; do not treat it as a pure cross-reference and mark it empty. 有? (have?) and garbled text cannot be reliably interpreted and are unresolved. These are project examples, not ESCO rules for Chinese boundaries.

R20 Uses of model review and human confirmation

Related examples: [EX113](#) (Appendix A)

Silver may use labels generated by a teacher model and authorized model-review results as training labels without human confirmation of every record. Model review must still give reasoned scope, boundary, and type decisions and pass structural checks. Merely changing adjudication_required to completed in bulk is insufficient.

Model-resolved ambiguities may enter the pool of Silver annotations that have passed the required quality checks; isolate whole records still unresolved. Passing these quality checks does not by itself make a record eligible for training: check duplicates, shared sources, and test separation separately. Record human-review status, model-review status, original generating model, and version separately. A teacher's self-review is not an independent second annotator and does not support a new independent blind IAA estimate. [LK77]

Each review records at least the ID, source, old/new labels, scope/boundary/type issue, rationale, rule ID, review method, model-identity source, and version. Use verifiable model-run identifiers; record unknown when unverified.

The historical 53 model-resolved records and three groups of identical-text conflicts are preserved in the model case log, with confirmation method recorded as model review. Reassess the same terms in new contexts. Save later human changes as a new version and retain the previous one.

R21 Scope evidence and preservation of local meaning

Related examples: [EX114](#)–[EX116](#) (Appendix A)

Use only the current sentence in this version, including its component clauses. Do not consult preceding or following sentences, the job title, or the full advertisement to supply missing meaning. Record IDs and queue metadata serve only to locate records. If the sentence does not establish annotation scope, request review. Full historical advertisements remain available for provenance, not as additional context for current decisions.

Usually exclude introductory words such as 熟悉 / 具备 (be familiar with/possess), but retain heads expressing real occupational actions. If removing them leaves content unable to express the original requirement, retain the necessary action in the current duty context and record why.

Assess category terms and specific names separately. Do not add an entity for a generic category that merely introduces a list. Where the sentence independently requires category-level knowledge or work, it may be annotated alongside non-overlapping specifics. Here 常用算法编程语言, 如 R, Python, C 等 (programming languages commonly used to implement algorithms, such as R, Python, C) only lists specific languages; this does not justify deleting all database or framework categories. Request review if independent requirement status remains unclear. [Project rule]

R22 Data states and experimental versions

Difficult and easy identify the original conflict and non-conflict queues, not human difficulty ratings. The absence of an ambiguity flag in a historical record does not mean that the record is unambiguous. Preserve source-queue status, historical ambiguity, latest review decision, empty status, duplicate group, and training-inclusion status separately.

Preserve source text, IDs, original batches, annotation-generation procedures, and historical A100/Gold150 versions. For identical text with different labels, first compare permitted context and handbook versions. Different conditions do not prove an error. For disagreements under the same conditions, review and record the chosen labels, rules, and reasons. A separate training copy may be deduplicated while original records remain. ID prefixes help retrieve provenance but do not prove a shared source document; disclose this limitation when separating data by source.

Gold150 must not be used for training, early stopping, hyperparameter selection, or test-feedback-based prompt changes. Disclose any involvement in historical rule development. Create and freeze a development set from non-test data using source-group splits before training. New Silver sentences originate from the

2601-record collection; a model trained on them cannot use the same collection as an independent main test set or be compared directly with historical main-table F1.

Historical dataset counts and the development-set plan at that time appear under D10 in Appendix B. They do not describe the current dataset size or the present status of the split.

R23 Splitting coordinated items under shared experience context

Related examples: [EX117](#)–[EX124](#) (Appendix A)

When items joined by 或 (or), 和 (and), or 与 (and) name independent activities, tools, or other objects of practical work, check whether a shared relevant-work-experience phrase in the same sentence applies to all of them. If it does and splitting preserves their meaning, annotate separate continuous source spans. The shared experience phrase may remain outside the spans while guiding their interpretation. Classify each item using the practical-experience requirement in the sentence.

First establish the shared phrase's local scope, then test each item's independent identifiability and unchanged type. Select shorter independent spans when these conditions hold. Retain the shortest complete long span if removal leaves a category unable to express the requirement or loses necessary action, object, experience, or willingness. Register non-unique answers for review instead of splitting mechanically at conjunctions.

Confirmed case 194

Record ID: 1838-s0008. The source follows. Offsets use Unicode code points, zero-based indexing, and an end position that is not included.

4、三年以上大中型互联网公司经营或 BI 相关工作经验，互联网公司广告运营分析优先；

English gloss: 4. At least three years of relevant work experience in internet company operations or BI at medium-sized or large companies; experience in advertising operations analysis at internet companies preferred.

This version uses three S spans:

[9,16) → 互联网公司经营 → S。

English gloss: [9,16) → internet company operations → S.

Reading the interval: [9,16) selects seven characters from source position 9 up to, but not including, position 16, yielding 互联网公司经营 (internet company operations). Select the source accurately; do not rewrite it to fit an existing interval.

[17,19) → BI → S。

[26,37) → 互联网公司广告运营分析 → S，此项不变。

English gloss: [26,37) → advertising operations analysis at internet companies → S; unchanged.

The first two items are joined by 或 (or) and share the following phrase 相关工作经验 (relevant work experience). The sentence requires practical experience, so both are S in this context. Current guidance selects 互联网公司经营 (internet company operations) and BI, leaving the shared experience phrase in context. BI 相关工作经验 is indeed a continuous source substring, at [17,25); choosing BI alone is a boundary decision, not a claim that the longer phrase is absent. Do not also retain a long span covering both items.

Revision scope and historical records

Historical A100 retains 互联网公司经营或 BI 相关工作经验 (relevant experience in internet company operations or BI), S, [9,25), and the unchanged [26,37). R23 shows the three shorter spans adopted later. The two solutions belong to their respective versions. Historical data, entity counts, and agreement statistics are not recalculated by this change. Save data adopting the new guidance as a new version with documented differences.

Revision scope: that revision adopted only the case-194 split; four other discussed differences retained their original statuses. Exact Chinese boundaries are project rules.

Appendix A Complete case collection and index

This appendix contains 124 EX reference entries; 23 numbered cases also include full source text and answers. Reference markers link to the references at the end, and explanations correspond to the main rules.

Base handbook version: B.sop_v4.2.14. The case appendix was first compiled on 2026-09-29; this document was revised on 2026-10-07.

The appendix follows the main handbook and R08–R23, collecting sentences, short examples, counterexamples, conditional illustrations, and numbered cases for consultation.

Instructions for use

1. EX numbers identify reference entries; case numbers such as 102, 128, and 185 retain the original numbering. One EX entry may contain several contrasting examples. Count entries and original records separately.
 2. Part I follows the handbook's rule order. Part II expands the 23 numbered cases explicitly cited in the main text with full source sentences and labels. Repeated references are index entries, not additional samples.
 3. Teaching fragments are not necessarily complete advertisement sentences. Preserve ellipses, slashes, conditions, and alternative boundaries from the source handbook. Do not invent record IDs or offsets for them.
 4. Unresolved does not mean empty. Provisional S is not final S. Examples with multiple candidate boundaries are identified explicitly. Apply scope, continuous complete boundaries, local type, and review status in that order.
 5. Numbered cases mainly derive from historical Silver-plus model-assisted annotation and review/authorized re-review; interpret them under that provenance. Case 194 displays current R23 guidance while preserving its historical-boundary explanation.
 6. Coverage includes the positive, negative, and conditional examples explicitly given from the label definitions through R23. Other records in linked logs and later proposals are maintained through a separate revision process.
- Reference markers such as [\[Z22\]](#) and [\[EQF\]](#) follow the main handbook; full references are at the end. [Project rule] indicates an annotation rule established for this project, not an external reference. A citation does not mean that the cited work prescribes the exact Chinese boundaries.

Handbook section index

Main handbook section	Appendix entries	Notes
Four labels and their evaluation grouping	EX001 to EX004	4 entries, including repeated references and contrast groups
Order of decisions when resolving ambiguity	EX005	1 entry, including repeated references and contrast groups

Main handbook section	Appendix entries	Notes
Education and qualification names labeled K	EX006 to EX008	3 entries, including repeated references and contrast groups
Communication and reporting annotated as T	EX009 to EX013	5 entries, including repeated references and contrast groups
Occupational abilities and general competences	EX014 to EX017	4 entries, including repeated references and contrast groups
Technical domains in context	EX018 to EX021	4 entries, including repeated references and contrast groups
Key distinctions	EX022 to EX032	11 entries, including repeated references and contrast groups
Boundaries for background and experience	EX033 to EX037	5 entries, including repeated references and contrast groups
When to retain the colloquial word 会	EX038 to EX040	3 entries, including repeated references and contrast groups
Selecting source-text spans	EX041 to EX049	9 entries, including repeated references and contrast groups
Tool names as S and knowledge expressions as K	EX050 to EX057	8 entries, including repeated references and contrast groups
Additional guidance on scope and complete expressions	EX058 to EX064	7 entries, including repeated references and contrast groups
Contrasting examples	EX065 to EX076	12 entries, including repeated references and contrast groups
Supplementary rules R08 to R13	EX077 to EX078	2 entries, including repeated references and contrast groups

Main handbook section	Appendix entries	Notes
R09 General problem solving and practical methods	EX079 to EX080	2 entries, including repeated references and contrast groups
R10 Additional cases and review status	EX081 to EX085	5 entries, including repeated references and contrast groups
R11 Unresolved scope boundaries and types	EX086 to EX088	3 entries, including repeated references and contrast groups
R12 Model identity and execution records	No separate case entry	Process or version-management rule; no invented example.
R13 Character positions and source protection	No separate case entry	Process or version-management rule; no invented example.
R14 Meaning preserving splits and necessary long spans	EX089 to EX092	4 entries, including repeated references and contrast groups
R15 Knowledge methods tools and distinct requirements within a sentence	EX093 to EX097	5 entries, including repeated references and contrast groups
R16 Willingness dispositions and research ability	EX098 to EX101	4 entries, including repeated references and contrast groups
R17 Occupational functions and general coordination	EX102 to EX107	6 entries, including repeated references and contrast groups
R18 Review records and data versions	EX108	1 entry, including repeated references and contrast groups
R19 Separate empty and difficult records	EX109 to EX112	4 entries, including repeated references and contrast groups
R20 Uses of model review and human confirmation	EX113	1 entry, including repeated references and contrast groups

Main handbook section	Appendix entries	Notes
R21 Scope evidence and preservation of local meaning	EX114 to EX116	3 entries, including repeated references and contrast groups
R22 Data states and experimental versions	No separate case entry	Process or version-management rule; no invented example.
R23 Splitting coordinated items under shared experience context	EX117 to EX124	8 entries, including repeated references and contrast groups

Part I Entries in handbook order

Four labels and their evaluation grouping

EX001 Education majors and non-language qualifications

[Return to the corresponding main rule](#)

手册词例：本科及以上学历、计算机专业、ISO 27001、PMP。

English gloss: Handbook examples: bachelor's degree or higher; computer science major; ISO 27001; PMP.

Label: K. Applies to domain facts/principles/theories, core domain names in explicit industry-background/experience requirements (project broad K), and education/non-language occupational qualification names (K under project rules; no separate qualification label).

Reading note: assigning education and non-language qualifications to K is a project rule. These examples are not word-by-word Chinese annotations provided by ESCO.

EX002 Occupational activities and tools

[Return to the corresponding main rule](#)

手册词例：维护、测试、接口对接、岗位运用中的 shell。

English gloss: Handbook examples: maintenance; testing; interface integration; shell in occupational use.

Label: S. Applies to the use of knowledge through tools, methods, and specific occupational activities.

Reading note: the tools must occur in a context requiring occupational use. These short phrases lack complete source sentences and do not establish fixed labels for every occurrence of the same terms.

EX003 Communication cooperation and responsibility

[Return to the corresponding main rule](#)

手册词例：沟通能力、沟通管理、客户汇报、责任心。

English gloss: Handbook examples: communication skills; communication management; reporting to clients; sense of responsibility.

Label: T. Applies to transversal soft skills and dispositions. Appearing in a duty statement does not change this classification.

EX004 Natural languages and language qualifications

[Return to the corresponding main rule](#)

手册词例：英语、英语六级、CET-6、日语 N2、英文阅读能力。

English gloss: Handbook examples: English; College English Test Band 6; CET-6; Japanese proficiency at N2 level; English reading ability.

Label: L. Language names, proficiency, examinations, and certificates remain complete L spans.

For the L/T distinction, see Communication and reporting. Natural-language proficiency is L; general interpersonal communication is T. The phrase 英语沟通 (communication in English) alone does not determine a unique type without the complete requirement.

Reading note: language means natural language, not programming language.

Reviewing scope boundaries and type

EX005 Status when type remains uncertain

Establish scope and boundaries, then compare candidate types. Use provisional S and request review only if type alone remains unresolved.

[Return to the corresponding main rule](#)

Handbook example and treatment

Only when scope and boundaries are established, but comparison of the candidate types still leaves the type uncertain, may S be recorded provisionally with adjudication_required (review required). Do not include it in the final human reference (Gold) labels or use it for training before review. Provisional S cannot resolve uncertainty about whether to annotate; see R11 for scope and boundary issues. [\[Z22\]](#) [Project rule]

Reading note: provisional S is permitted only after scope and boundaries are clear. It remains pending review, not a final training label.

Education and qualification names labeled K

EX006 Alternative education expressions

[Return to the corresponding main rule](#)

Handbook example and treatment

“本科及以上学历” (bachelor's degree or higher), “大学本科及以上” (university bachelor's degree or higher), and “Bachelor Degree” are K. [Project rule] SkillSpan Appendix B.2, item 8, labels Bachelor Degree as KNOWLEDGE and the following subject names separately. [\[Z22\]](#)

EX007 Employment duration versus education qualifications

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

Select all of 本科及以上学历 (bachelor's degree or higher) as K. 5 年以上 (five years or more) only indicates employment duration and is not a standalone entity. Treat them separately; their shared 以上 (above/more than) does not justify excluding both. [Project rule]

EX008 Do not truncate education thresholds

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

Retain threshold wording such as 及以上 / 含以上 (or above/including higher levels). Select 本科及以上学历 (bachelor's degree or higher), not just 本科 (bachelor's level). This is a project boundary rule. [Project rule]

Communication and reporting annotated as T

EX009 Four communication and reporting examples

[Return to the corresponding main rule](#)

Handbook example and treatment

沟通能力 (communication skills), 沟通管理 (communication management), and 客户汇报 (reporting to clients) → T. For expressions involving a specific natural language, apply the main L/T conditions; do not assign English communication a fixed T. [Project rule]

EX010 Conceptual source of report facts

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

ESCO places report facts in communication category T4.1 and identifies it as reusable across contexts. T4.1 explains the conceptual source; it is not an additional annotation label. [\[ESCO-T\]](#)

Reading note: this entry explains the conceptual source; the Chinese boundary of 客户汇报 (reporting to clients) follows project rules.

EX011 Scope of the English soft-skill references

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

SkillSpan has no separate T label, and its SKILL class is not identical to this project's S. General communication and related dispositions are T here and join SKILL only in two-class evaluation. [\[Z22\]](#) [\[Say18\]](#) [Project rule]

Reading note: SkillSpan SKILL is not the standalone S of this four-class scheme. Do not infer a new LSKT label for customer service from this reference note.

EX012 Duty framing does not change general competence

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

General communication remains T when written as 负责.....工作 (responsible for ... work). Assess specific technical operations as S. [\[ESCO-T\]](#) [Project rule]

EX013 Technical interfaces and interpersonal coordination

[Return to the corresponding main rule](#)

Handbook example and treatment

Occupational technical actions are S: 接口对接 (interface integration), 信息梳理 (organizing information with a specific business object), and problem handling with a necessary object. The word 对接 (liaise/interface) alone does not make interpersonal coordination S. 协调外部资源 (coordinate external resources) is a complete T, retaining the necessary object. [\[Z22\]](#) [Project rule]

Reading note: problem handling plus necessary object is a conditional illustration, not a continuous source span that can be output directly.

Occupational abilities and general competences

EX014 General versus occupational abilities

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

Distinguish S and T by specific occupational activity versus general ability. Apply the main research criteria using activity, method, and task evidence rather than mandatory keywords. [Project rule]

EX015 Three occupational abilities

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

1. Occupational dependence: does the requirement explicitly depend on a tool, method used to perform a specific occupational task, or occupational object? If so, assess S; examples include 编码能力 (coding skills), 产品设计能力 (product-design ability), and 数据分析能力 (data-analysis ability).

EX016 General cognitive and interpersonal abilities

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

2. General competence: does the expression mainly concern general cognition, self-management, communication, or interpersonal ability? If so, assess T; examples include 沟通能力 (communication skills), 学习能力 (learning ability), and 沟通管理 (communication management). Use across several industries alone does not establish T; specific occupational functions such as project management still follow R17.

EX017 Conditional rules for research ability

[Return to the corresponding main rule](#)

Handbook example and treatment

3. Classify research ability by the requirement expressed. General inquiry, thinking, or research ability is T. Specific occupational research activities, use of research methods, or identifiable research tasks are assessed as S. “论文 / 实验 / 课题” (paper/experiment/research project) are possible evidence, not mandatory words; their presence alone also does not establish S. Requirements limited to theory or methodological knowledge are K. If knowledge, practice, and general ability cannot be distinguished, record alternatives and request review. [Project rule]

Technical domains in context

EX018 安全 and 网络 in majors or certifications

[Return to the corresponding main rule](#)

Context: a discipline, certification, or major name. Condition: an explicit preceding list of majors, certifications, disciplines, or education requirements.

Handbook decision and examples: K. 网络安全专业 (a network-security major); 安全 / 网络 (security / networks) in the certification list.

Reading note: 安全 / 网络 (security/networks) depends on the certification-list context. These words are not always K.

EX019 Networks and ICT as operational objects

[Return to the corresponding main rule](#)

Context: objects of planning, operations/maintenance, or implementation. Condition: planning, maintenance, management, implementation, or integration follows, or the job involves that work.

Handbook decision and examples: S. In 规划网络及 ICT...运维计划 (plan network and ICT ... operations and maintenance plans), 网络 (network) and ICT 运维计划 (ICT operations and maintenance plans) are separate S spans.

Reading note: the ellipsis marks a schematic construction, not a complete source sentence. Do not invent offsets or generalize that all system objects must be separately annotated.

EX020 Excluding mere domain modifiers

[Return to the corresponding main rule](#)

Context: a generic domain modifier. Examples: 网络行业 (network industry), 安全领域 (security field), ICT 市场 (ICT market), with no independent competence or explicit background/experience requirement.

Handbook decision: no annotation. See the next section for the exception involving explicit industry background or experience requirements.

EX021 Guaranteeing or ensuring safety

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

Do not annotate “保证安全 / 确保安全” (guarantee/ensure safety) as an ability when it only states a goal or outcome. Where a specific safety operation or qualification name is expressed, classify the actual content; the shared word “安全” (safety/security) does not require a shared label.

Key distinctions

EX022 Reference note on participation and contribute

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

Follow the main rule for activities versus outcomes. Consider S when the current sentence identifies specific work the applicant is expected to perform. Do not separately annotate a desired state or metric alone. Verbs such as 优化 (optimize) and 提高 (improve) do not determine the answer. [Project rule]

Reading note: this is guidance on the cited literature, not a context-free blacklist for participation or contribute.

Source location: main handbook, Key distinctions; historical Markdown source, line 86.

EX023 Analyzing the effectiveness of promotional activities

[Return to the corresponding main rule](#)

手册原例：分析推广效果

English gloss: Source example: analyze the effectiveness of promotional activities.

Treatment: an activity or method.

Decision: S.

Source location: main handbook, Key distinctions; historical Markdown source, line 93.

EX024 Developing promotional strategies

[Return to the corresponding main rule](#)

手册原例：制定有效的推广策略

English gloss: Source example: develop effective promotional strategies.

Treatment: an activity or method.

Decision: S.

Source location: main handbook, Key distinctions; historical Markdown source, line 94.

EX025 Optimizing exposure and the conversion rate

[Return to the corresponding main rule](#)

手册原例：优化曝光与转化率

English gloss: Source example: optimize exposure and the conversion rate.

When the example is an advertising-optimization duty, retain 优化曝光与转化率 (optimize exposure and the conversion rate) as one S. Exposure and conversion rate alone lose the activity. If the same wording merely describes a target or achieved result, do not extract it solely on this example's authority.

Decision: one S retaining the optimization action and its objects.

Source location: main handbook, Key distinctions; historical Markdown source, line 95.

EX026 Guaranteeing traffic and growth

[Return to the corresponding main rule](#)

手册原例：保证流量/用户增长

English gloss: Source example: ensure traffic/user growth.

Treatment: a desired outcome or KPI.

Decision: do not annotate.

Source location: main handbook, Key distinctions; historical Markdown source, line 96.

EX027 Meeting channel targets

[Return to the corresponding main rule](#)

手册原例：确保渠道目标达成

English gloss: Source example: ensure channel targets are met.

Treatment: a desired outcome.

Decision: do not annotate.

Source location: main handbook, Key distinctions; historical Markdown source, line 97.

EX028 Outcome guarantees versus occupational quality assurance

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

Do not annotate “保证 / 确保” (guarantee/ensure) when these merely promise a result. Check the following text for independent competences. Assess “质量保证” (quality assurance) as a whole when it names an occupational activity; do not mechanically remove “保证”. See R17. [\[PB05\]](#) [Project rule]

Reading note: occupational quality assurance (QA) differs from an outcome clause promising that quality standards will be met. See case 124 for the complete occupational example.

EX029 Pure outcomes and system states

[Return to the corresponding main rule](#)

手册例子：增长、流量、目标、业绩、稳定、质量、满意度、达标、合规、可靠、高效、优质；保证系统稳定；确保服务质量；保证安全

English gloss: Examples: growth, traffic, targets, performance, stability, quality, satisfaction, meeting standards, compliance, reliability, efficiency, high quality; guarantee system stability; ensure service quality; guarantee safety.

Condition: pure outcome/state nouns. Treatment: do not annotate the phrase.

Reading note: this table concerns pure outcomes/states. Do not remove quality or safety when they are necessary objects within a technical activity.

EX030 Data backup as a desired outcome

“确保数据备份策略有效” (ensure the data-backup strategy is effective) states a desired outcome without an explicit knowledge or operational requirement. Extract no entity in this example.

[Return to the corresponding main rule](#)

Example: 确保数据备份策略有效 (ensure the data-backup strategy is effective) → no entity.

For this example: the sentence states only a desired outcome, so no entity is extracted.

Reading note: this decision applies to this sentence. For other sentences containing ensure, assess the content; do not exclude an explicit occupational activity merely because of that verb.

EX031 Incident response

[Return to the corresponding main rule](#)

Example: 保证按 SLA 完成故障响应 (ensure incident response is completed within the SLA) → 故障响应 (incident response), S.

Condition: action plus outcome. Treatment: annotate the action only.

EX032 Analysis development optimization and metric objects

[Return to the corresponding main rule](#)

Handbook example and treatment

Metrics following 分析 / 制定 / 优化 (analyze / formulate / optimize) may be the objects of an activity and support S. A preceding 保证 (ensure) does not automatically make the entire sentence empty or every item S.

Boundaries for background and experience

EX033 Generic experience

[Return to the corresponding main rule](#)

手册例子：工作经验、项目经验、丰富经验

English gloss: Examples: work experience; project experience; extensive experience.

Condition: generic experience without a domain or activity. Treatment: do not annotate.

EX034 Core domain names in industry experience

[Return to the corresponding main rule](#)

Example: 有银行、咨询、服务行业工作经验优先考虑 (experience in banking, consulting, or the service industry is preferred) → 银行 (banking), K; 咨询 (consulting), K; 服务行业 (service industry), K.

Condition: an explicit industry-background/work-experience requirement. Treatment: each core domain name is K; exclude generic experience/background wording, duration, and preference conditions.

EX035 Activities that remain complete without 经验

[Return to the corresponding main rule](#)

手册例子: 大项目售前、数据分析、大数据开发、大数据项目实施、BI 项目实施

English gloss: Examples: presales work on large projects; data analysis; big-data development; big-data project implementation; BI project implementation.

Condition: a specific activity remains complete after removing experience. Treatment: activity phrase S, excluding experience.

Reading note: these fragments already exclude experience; do not invent complete recruitment sentences for them.

EX036 Product types that require 经验

[Return to the corresponding main rule](#)

手册例子: 产品经验、内容型产品经验

English gloss: Examples: product experience; experience with content-based products.

Condition: experience in a specific function or product type, where removing 经验 (experience) leaves only a category and changes the meaning. Retain 经验 and label the full span S.

EX037 Limits of using industry background to represent knowledge

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

When the applicant is explicitly required to have industry background, assigning the core domain name to broad K is a project rule for representing background knowledge. First confirm that an applicant-background requirement is actually expressed. Company activities, the job's industry, product markets, personal interests, and user identities do not by themselves establish such a requirement. Retain “行业” (industry) in “服务行业” (service industry) to make the domain meaning clear.

When to retain the colloquial word 会

EX038 Four colloquial interpersonal competences

[Return to the corresponding main rule](#)

手册原例: 会聊天、会说话、会来事、会做人

English gloss: Source examples: can hold a conversation; know how to speak appropriately; be socially adept; know how to get along with people.

Handbook decision: retain the whole T span, including 会 (know how to).

Reading note: 会聊天, 会说话, 会来事, and 会做人 are four separate T spans. Complete phrase means each colloquial expression, not all four concatenated into one span.

EX039 Using tools and performing tasks with 会

[Return to the corresponding main rule](#)

手册原例: 会使用 Excel、会修电脑、会写 SQL

English gloss: Source examples: able to use Excel; know how to repair computers; know how to write SQL.

Confirmed annotation: Excel S; 修电脑 (repair computers) S; 写 SQL (write SQL) S.

会使用 Excel (able to use Excel) states generic tool use; the confirmed span is Excel. For 会修电脑 (can repair computers) and 会写 SQL (can write SQL), retain the repair and writing actions but exclude 会. If a sentence specifies using a tool to perform a task, assess the full task; this example does not justify removing 使用 (use) in every case.

EX040 Examples of excluding triggers

[Return to the corresponding main rule](#)

Handbook example and treatment

For 善于 / 熟悉 / 掌握 / 了解 / 精通 / 具备 / 有 (be good at / be familiar with / have a command of / understand / be proficient in / possess / have), select the following content: 善于与人沟通 → 与人沟通 (communicate with others), T; 有服务意识 → 服务意识 (service-oriented mindset), T. [Z22] Appendix B.3.6 treats advanced knowledge of and proficient in as triggers outside the span.

Reading note: 与人沟通 (communicate with others), T; 服务意识 (service-oriented mindset), T. Advanced knowledge of and proficient in illustrate introductory triggers, not target entities.

Selecting source-text spans

EX041 Truncated words are invalid boundaries

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

1. Select continuous, complete source substrings without correcting spelling or adding words. “支持服” and “操作系统的问” are incorrect because they truncate words (fragments of support service and operating-system problem). The technical check is sentence[start:end] == span: slicing the recorded positions must reproduce the entity text exactly. [Z22] [D18] [AP08] [Project rule]

Reading note: 支持服 and 操作系统的问 are deliberately truncated counterexamples, not valid entities to import.

EX042 Short complete examples

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

2. Select the shortest complete span that preserves meaning. Semantic completeness determines length. Python, 计算机专业 (computer science major), and 沟通能力 (communication skills) can each stand alone. Annotate independent duties separately, retaining each complete meaning. [Z22] [AP08] [Kr95] [Project rule]

EX043 Coordinated splitting and shared actions

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

3. Split items joined by 、 (enumeration comma), 和 / 与 (and), or 或 (or) only when each independently expresses the original requirement. Retain a complete span when splitting would cut a word, remove necessary shared wording, or change the meaning. For example, 优化曝光与转化率 (optimize exposure and the conversion rate) cannot be reduced to two metric names: doing so loses the required action 优化 (optimize). [Z22] [UD20] [Project rule]

EX044 Triggers and the 会聊天 exception

[Return to the corresponding main rule](#)

Handbook example and treatment

4. Exclude introductory “熟悉 / 掌握 / 了解 / 精通 / 具备” (be familiar with/have command of/understand/be proficient in/possess), and select their content. [Z22] [Project rule] Colloquial interpersonal expressions such as “会聊天” (can hold a conversation) are exceptions; see the preceding section. [Project rule]

EX045 Python language principles versus tool use

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

5. Classify programming languages, frameworks, software, and tools by context. A tool name in occupational use may be S. Explicit course content, principles, fundamentals, or syntax are K as complete knowledge phrases, such as “Python 语言原理” (principles of the Python language). If scope and boundaries are clear but type is unresolved, record a provisional candidate and request review. [EQF] [ESCO14] [Z22] [Project rule]

EX046 Language examinations and non-language certifications

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

6. Natural-language names, proficiency, and examinations, such as 英语六级 (College English Test Band 6), CET-6, and 日语 N2 (Japanese proficiency at N2 level), are L. Non-language technical or occupational qualifications such as ISO and OCJP are K under project rules. Older versions used different labels; do not overwrite archived data on that basis. [ESCO-L] [ESCO-Q] [Project rule]

Reading note: ISO and OCJP are K in non-language qualification contexts. Retain the complete certification name when the source supplies it.

EX047 Applicant procedures and recruitment work

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

7. Applicant-facing procedures such as application, medical examination, public announcement, and eligibility checks are not annotated, nor are benefits, statutory insurance/housing-fund benefits, shifts, locations, or slogans in themselves. Record spans: [] only after confirming that the whole sentence has no target entity. Insufficient evidence, missing context, and disagreement are unresolved states; see R19. Recruitment,

interviewing, and background checks performed by recruitment staff may be occupational activities, so do not exclude them by keyword alone. [\[Z22\]](#) [Project rule]

Reading note: these are scope categories, not a list that always excludes words such as registration and interviews. When a recruiter performs those activities, assess them as possible occupational work.

EX048 Appearance health duration and driving licences

[Return to the corresponding main rule](#)

Handbook example and treatment

8. Exclude appearance, physical health, numerical durations (retain the ability itself), and driving licences for non-driver roles. [\[Z22\]](#) [Project rule]

EX049 Two contexts for SQL and Python

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

10. The same name can receive different types in different contexts. Assess SQL/Python used in actual work as tool-name S. When the requirement concerns courses, principles, fundamentals, or syntax, annotate the complete knowledge phrase as K rather than the tool name alone. See the following section. [\[ESCO14\]](#) [Project rule]

Tool names as S and knowledge expressions as K

EX050 Classifying listed examples from local context

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

See Tool names as S and knowledge expressions as K. Find source evidence of use, practice, or knowledge. Request review if familiarity or the job title alone is insufficient. 如 (such as) does not determine a listed item's type. [Project rule] [EQF]

EX051 Programming-language tools in a list

[Return to the corresponding main rule](#)

手册原例：掌握常用算法编程语言，如 R, Python, C 等

English gloss: Source example: have a command of programming languages such as R, Python, and C that are commonly used to implement algorithms.

Confirmed annotation: R S; Python S; C S. Exclude 掌握 (have command of). Here 常用算法编程语言 (programming languages commonly used to implement algorithms) only introduces the list and is not separately annotated.

Common error: changing all examples to K or nesting K inside a long S.

The category only introduces the list here. If another sentence independently requires category-level knowledge or work, reassess it under R21 rather than deleting it automatically.

EX052 Office software in a tool list

[Return to the corresponding main rule](#)

手册原例：熟悉使用 Word，Excel，PPT

English gloss: Source example: be familiar with using Word, Excel, and PowerPoint.

Handbook decision: Word, S; Excel, S; PPT, S.

Common error: including 熟悉使用 (be familiar with using).

EX053 Python and data structures in courses

[Return to the corresponding main rule](#)

手册原例：计算机专业课程包括 Python、数据结构

English gloss: Source example: computer science courses include Python and data structures.

Type illustration: Python and 数据结构 (data structures) are assessed as K in an established applicant course-knowledge requirement. This fragment is not a complete requirement and does not independently establish 计算机专业 (computer science major) as an entity.

Common error: labeling Python as S.

Reading note: check whether the complete sentence requires the applicant to possess this knowledge. A course or textbook description alone is not an extraction case. 包括 (include) alone does not determine scope or type.

EX054 Python principles and interpreter implementation

[Return to the corresponding main rule](#)

手册原例：了解 Python 语言原理 / 解释器实现

English gloss: Source example: understand Python language principles / interpreter implementation.

Handbook decision: Python 语言原理 (principles of the Python language), K; 解释器实现 (interpreter implementation), K.

Common error: selecting only Python and omitting the necessary knowledge qualifier in the source.

EX055 Basic Python knowledge

[Return to the corresponding main rule](#)

手册原例：具备 Python 基础知识（后文不再要求开发）

English gloss: Source example: have basic Python knowledge, without a subsequent development requirement.

Handbook decision: Python 基础知识 (basic knowledge of Python), K.

Common errors: selecting only Python; including 具备 (possess).

Reading note: if the source says Python 基础知识, retain the space after Python; do not change it to Python 基础知识.

EX056 Conditional illustration for Python syntax and the standard library

[Return to the corresponding main rule](#)

手册原例：教材：Python 语法与标准库

English gloss: Source example: textbook: Python syntax and the standard library.

Confirmed decision: in an applicant knowledge requirement, annotate “Python 语法与标准库” (Python syntax and the standard library) as one K span. Python qualifies both items. Retain the full phrase so the library remains identifiable, without inserting words absent from the source.

Common error: selecting only Python and omitting the necessary knowledge qualifier in the source.

Reading note: “教材：Python 语法与标准库” (textbook: Python syntax and the standard library) describes textbook content, not an applicant requirement. The K answer above applies only in an established knowledge-requirement context. Request review if scope is unclear.

EX057 Principles of SQL and database querying

[Return to the corresponding main rule](#)

手册原例：了解 SQL 原理 / 用 SQL 查库

English gloss: Source example: understand the principles of SQL / use SQL to query a database.

Confirmed decision: “SQL 原理” (principles of SQL) is K; “SQL 查库” (querying a database with SQL) is S. Retain 查库 because it names the specific operation. Exclude the introductory 用 (using).

Common error: combining examples from two contexts into one entity.

Reading note: the slash separates knowledge from operation. Preserve the source space in “SQL 查库”; do not rewrite it as “SQL 查库”.

Additional guidance on scope and complete expressions

EX058 Company promotion benefits and applicant requirements

[Return to the corresponding main rule](#)

Handbook example and treatment

1. Establish scope before type. Check whether the expression concerns an applicant's competence, knowledge, or qualification before selecting boundaries and classifying. Big data or cloud computing listed as company technical advantages are not therefore K. Do not extract 网络 (networks) from 网络课程 (online courses) mentioned as a reimbursement benefit. Company encouragement of self-drive is not automatically an applicant requirement. When duties and benefits coexist, assess each clause rather than clearing the whole sentence.

Reading note: company technical advantages, online-course benefits, and encouragement of self-drive are not automatically applicant requirements. Assess independent duties in the same sentence separately.

EX059 Debugging methods principles and standards

[Return to the corresponding main rule](#)

Handbook example and treatment

2. Methods are not always K. Technical methods used to perform occupational tasks are assessed as S, while theories, principles, and standards are K: Linux 内核调试方法 (Linux kernel debugging methods), S; Linux 系统原理 (Linux system principles), K; 编码规范 (coding standards), K. Exclude familiar with and common, but do not remove 方法 (method) merely to shorten the span if this changes its object.

EX060 Necessary objects disposition qualifiers and system properties

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

3. Completeness takes priority over brevity. Do not shorten 协调外部资源 (coordinate external resources), T, to 协调 (coordinate), or remove necessary qualifiers from 严谨的工作态度 (a careful and meticulous approach to work), T. When product documentation is the object of a writing activity, do not add nested K. System stability is a system property, not a person's T.

EX061 Shared structure in expression ability and KVM

[Return to the corresponding main rule](#)

Handbook example and treatment

4. Distinguish independently meaningful items joined by and/or from items that need shared wording. Annotate independently identifiable tools separately. Keep one span when shared wording is necessary, as in 口头、书面表达能力 (oral and written expression ability), T, and KVM 的开发和实施 (KVM development and implementation), S. Do not split mechanically at and, or combine independent duties into one S. Complex lists with omitted shared words still require review; see the issue register.

Reading note: the single KVM development/implementation S illustration coexists with the separable architecture-design/maintenance spans in case 120 under R17. The test is meaning loss, not the mere presence of a shared object.

EX062 No added words truncated characters or spelling changes

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

5. Select source text only: do not add words or truncate them. If the source only contains Memory, do not create Memory 开发经验 (Memory development experience). Do not cut 缓存 (cache) or RocketMQ mid-word. Preserve spelling, capitalization, and spaces. A normalized name may be recorded separately but cannot replace the selected source text.

Reading note: Memory 开发经验 is a prohibited candidate because it adds words absent from that source span. 缓/存 and Ro/cketMQ illustrate incorrect splits within words, not separate entities.

EX063 Category terms for frameworks databases and design tools

[Return to the corresponding main rule](#)

Handbook example and treatment

6. Category nouns do not automatically add K. Check whether category expressions such as frameworks, databases, and design tools independently express a knowledge requirement. Being a noun is insufficient. Distinguish practice from principles. Register uncertain category/tool boundaries instead of claiming that a uniform answer has already been established.

Reading note: nounhood or category status alone determines neither type nor separate annotation.

EX064 Reviewing candidate annotations

[Return to the corresponding main rule](#)

Handbook example and treatment

7. Candidate annotations require review. Model suggestions and alternatives in comparison records are not automatically human reference labels. Retain the source, candidate spans, confirmed spans and types, reasons, rule IDs, reviewer, and version. Keep undecided records pending. Corrections made after seeing model suggestions are not the original independent blind annotations used to calculate agreement. [Project rule]

Contrasting examples

EX065 Office-software contrast

[Return to the corresponding main rule](#)

手册原例：熟悉使用 Word，Excel，PPT 等办公软件

English gloss: Source example: familiar with using Word, Excel, PowerPoint, and other office software.

Handbook decision: Word, S; Excel, S; PPT, S.

Common errors: including 熟悉使用 (be familiar with using); combining the items into one long S span.

References and rule sources: [\[Z22\]](#) [Project rule]

EX066 Education and English-examination contrast

[Return to the corresponding main rule](#)

手册原例：本科及以上学历，大学英语 6 级

English gloss: Source example: bachelor's degree or higher; College English Test Band 6.

Handbook decision: 本科及以上学历 (a bachelor's degree or higher), K; 大学英语 6 级 (College English Test Band 6), L.

Common errors: omitting the education requirement; assigning K to CET-6.

References and rule sources: [\[Z22\]](#) [\[ESCO-Q\]](#) [\[ESCO-L\]](#)

EX067 Maintenance and support services

Boundary note: select 支持服务 (support services), S, as a complete expression.

[Return to the corresponding main rule](#)

手册原例：维护和支持服务

English gloss: Source example: maintenance and support services.

Treatment: 维护 (maintenance), S, plus 支持服务 (support services), S.

Common error: selecting the truncated 支持服 as S.

References and rule sources: [\[Z22\]](#) [Project rule]

EX068 Benefits as a confirmed-empty example

[Return to the corresponding main rule](#)

手册原例：五险一金，带薪年假

English gloss: Source example: five types of social insurance and a housing fund; paid annual leave.

Treatment: [].

Common error: forcing employee benefits into an entity label.

References and rule sources: [\[Z22\]](#)

EX069 Programming-language list contrast

[Return to the corresponding main rule](#)

手册原例：掌握...如 R, Python, C

English gloss: Source example: have command of ... such as R, Python, and C.

Handbook decision: R, Python, and C are separate S spans in an occupational-use context.

Common errors: assigning K to every example introduced by 如 (such as); selecting only the long category phrase.

References and rule sources: [Project rule]

EX070 Python-principles contrast

[Return to the corresponding main rule](#)

手册原例：了解 Python 语言原理

English gloss: Source example: understand Python language principles.

Handbook decision: Python 语言原理 (principles of the Python language), K.

Common error: selecting only Python.

References and rule sources: [\[ESCO14\]](#) [Project rule]

EX071 Promotional analysis versus growth outcomes

分析推广效果 (analyze the effectiveness of promotional activities), S; exclude 保证流量 / 用户增长 (ensure traffic/user growth) as an outcome.

[Return to the corresponding main rule](#)

手册原例：分析推广效果，保证流量/用户增长

English gloss: Source example: analyze the effectiveness of promotional activities and ensure traffic/user growth.

Treatment: 分析推广效果 (analyze the effectiveness of promotional activities), S; exclude the traffic/user-growth guarantee as an outcome. Develop ... strategies in the earlier explanation is a separate analogy, not extractable text from this sentence.

Common error: assigning S to 保证流量/用户增长 (ensure traffic/user growth).

References and rule sources: [\[Z22\]](#) [Project rule]

Reading note: develop ... strategies is a supplementary analogy absent from the source in this row. The actual sentence yields only 分析推广效果 (analyze the effectiveness of promotional activities), S; the growth outcome is excluded.

EX072 Coding communication and research contrast

[Return to the corresponding main rule](#)

手册原例：编码能力 / 沟通能力 / 科研能力

English gloss: Source example: coding skills / communication skills / research ability.

编码能力 (coding skills), S; 沟通能力 (communication skills), T; 科研能力 (research ability), T, only when it means general research ability. Assess specific research activities as S and methodological knowledge as K using the main rules.

Common error: treating every expression of being able to do something as S.

References and rule sources: [\[ESCO-T\]](#) [\[ONET\]](#) [Project rule]

EX073 Security and networks in contrasting contexts

Classify certification-list knowledge/qualifications as K and explicit network planning or maintenance as S. This illustrates activity context; actual boundaries depend on the complete source sentence.

[Return to the corresponding main rule](#)

手册原例：认证清单「安全」；规划网络...运维计划

English gloss: Source example: security in a certification list; planning networks ... operation/maintenance plans.

Treatment: certification-list knowledge/qualifications are K; network planning or maintenance is S. This is a contextual illustration, not permission to add ICT or other absent words.

Common error: assigning a fixed type to a term in every context.

References and rule sources: [\[Nav09\]](#) [Project rule]

Reading note: this contrast contains ellipses. ICT in the expanded illustration does not permit inserting characters or offsets into the source on the left.

EX074 Large-project presales experience

[Return to the corresponding main rule](#)

手册原例：大项目售前经验

English gloss: Source example: experience in presales work on large projects.

Handbook decision: 大项目售前 (presales work for major projects), S; exclude 经验 (experience).

Common error: retaining the whole phrase including 经验 (experience).

References and rule sources: [Project rule]

EX075 Communication and reporting duties

[Return to the corresponding main rule](#)

手册原例：负责与客户的沟通管理工作，...客户汇报工作

English gloss: Source example: responsible for managing communication with clients, ... and reporting to clients.

Handbook decision: 沟通管理 (communication management), T; 客户汇报 (reporting to clients), T.

Common error: assigning S merely because the sentence uses 负责 (be responsible for).

References and rule sources: [\[ESCO-T\]](#) [\[Z22\]](#)

EX076 Provisional recording and review of uncertain types

[Return to the corresponding main rule](#)

Illustration: the eligible source text and boundaries are confirmed, but candidate-type comparison does not resolve the type.

Treatment: record the candidate types and reasons and mark review as required. If provisional S is used under these annotation rules, the reviewer must still establish the final type; the temporary S is not a confirmed answer.

Order: establish scope and boundaries → compare candidate types → register unresolved status and review.

References and rule sources: [\[Z22\]](#) [\[AP08\]](#)

Reading note: maintain unresolved status and exclude the whole record from training until review finishes; see R11 and R19.

R08 Experience suffixes and source-text limits

EX077 Process-engineering experience and experience in robotics competitions

[Return to the corresponding main rule](#)

Handbook example and treatment

If the source explicitly requires specific practical experience and removing 经验 (experience) leaves only a category, domain, or competition name and changes the meaning, retain the complete expression: 工艺经验 (process-engineering experience) and 机器人竞赛经验 (experience in robotics competitions) → S. When a specific activity such as 数据分析 (data analysis) remains complete without experience, exclude the suffix. Apply the existing broad-K rule to industry background; exclude generic experience.

Reading note: retaining 经验 (experience) follows this project's boundary rule. It does not mean that all experience is S or that experience is equivalent to knowledge.

EX078 Do not add experience wording

[Return to the corresponding main rule](#)

Handbook example and treatment

Retaining words does not mean adding them. Do not add 经验 (experience) to a span unless it is part of the selected continuous source text. When coordinated items share a following experience phrase, first apply R14 and R23 to test whether coordinated items can be separated without meaning loss in local context. Retain the necessary shared span only if splitting changes the meaning; register unresolved boundaries. Interpret experience together with its activity or requirement.

R09 General problem solving and practical methods

EX079 General problem solving and analysis

[Return to the corresponding main rule](#)

Handbook example and treatment

解决问题 (solve problems) and 分析问题 (analyze problems) without a specific occupational object are T. For example, 善于解决问题和分析问题 yields two T spans, excluding 善于 (be good at). Consider S when a specific technical or business object is present. Do not reclassify all general abilities as S merely because the advertisement is for a technical job.

EX080 Embedded-software design methods and computer principles

[Return to the corresponding main rule](#)

Handbook example and treatment

In this example, 嵌入式软件设计方法 (embedded-software design methods) is S because it denotes methods used to perform specific occupational tasks in context; 计算机原理 (computer principles) in the same sentence is K. Keep introductory expressions such as 熟悉 (be familiar with), 了解 (understand), and 熟练 (proficiently) outside the entities. Determine the type from the content of the requirement.

R10 Additional cases and review status

EX081 Translating capabilities into business language

[Return to the corresponding main rule](#)

Handbook example and treatment

将公司技术能力/产品能力转化为商业语言 (translate the company's technical/product capabilities into business language) → one S. Technical and product capabilities are the objects of that activity, not nested K entities.

EX082 System outcomes enthusiasm and algorithm fundamentals

[Return to the corresponding main rule](#)

Handbook example and treatment

提高广告系统的用户体验、投放能力及变现效率 (improve advertising-system user experience, delivery capability, and monetization efficiency) is excluded in this pure-outcome context; assess actual improvement tasks under the main rules. 对解决挑战性问题充满热情 (be enthusiastic about solving challenging problems) is one T, without nested S. 数据结构和算法基础扎实 (have a solid foundation in data structures and algorithms) yields 数据结构 (data structures), K, and 算法基础 (algorithm fundamentals), K.

Reading note: this row contains three separate examples. A statement of system effects is not annotated; enthusiasm for solving challenging problems is T; 数据结构 (data structures) and 算法基础 (algorithm fundamentals) are K. Read them independently.

EX083 Repeated reference to embedded-software methods

[Return to the corresponding main rule](#)

Handbook example and treatment

3. 嵌入式软件设计方法 (embedded-software design methods) → S under R09; 熟练 (proficient) is not the deciding condition.

EX084 Production-transfer activities and goals

[Return to the corresponding main rule](#)

Handbook example and treatment

4. 构建新产品/新工艺转生产阶段转产流程 (establish the production-transfer process for new products/processes entering production) → one S. 承接公司新产品/新工艺的转产工作 (undertake production transfer for the company's new products/processes) → 新产品/新工艺的转产 (production transfer of new products/processes), S. Exclude 提升工艺团队转产能力 (improve the process team's production-transfer capability) if it is only a goal without an implementation method.

EX085 Conditional example under capability development

[Return to the corresponding main rule](#)

Handbook example and treatment

5. 能力提升: 媒体平台特性/互联网思维/运营推广方法论/接触前沿动态 (capability development: media-platform characteristics/internet thinking/operations and promotion methodology/exposure to current developments) requires context. Exclude confirmed benefits or post-employment prospects; assess each item if they are applicant requirements; otherwise register unresolved scope. A capability-development heading alone does not justify an empty annotation.

Reading note: exclude benefits/post-employment prospects; assess applicant requirements item by item; retain unresolved status when context is insufficient. The paragraph is not a fixed positive or negative example.

R11 Unresolved scope boundaries and types

EX086 Unresolved scope

[Return to the corresponding main rule](#)

Rule and examples consistent with the main text

Unresolved scope: retain established spans separately as review evidence and record disputed content and missing context. While any scope issue remains unresolved, do not mark the whole record as confirmed empty [], or send it into training merely because some entities are known. See R19.

Reading note: [] in a temporary format does not mean confirmed_empty when scope remains unresolved.

EX087 Unresolved boundaries

[Return to the corresponding main rule](#)

Handbook example and treatment

For unresolved boundaries, record a conservative, continuous, complete candidate and alternative boundaries with reasons. Do not put multiple overlapping answers into the final annotation layer.

Reading note: alternative boundaries belong in review records, not simultaneously in the final flat annotation.

EX088 Unresolved type

[Return to the corresponding main rule](#)

Handbook example and treatment

Unresolved type: after scope and boundaries are established, compare types in pairs. Use provisional S only if uncertainty remains, retaining alternative types and reasons.

Reading note: provisional S requires alternative types and unresolved status.

Source location: main handbook, R11 Unresolved scope, boundaries, and type; historical Markdown source, line 229.

R14 Meaning preserving splits and necessary long spans

EX089 Full source of case 130

[Return to the corresponding main rule](#)

Handbook example and treatment

案例 130 原句：5.熟悉 Linux 基本操作和原理，在云计算、深度学习等领域者有工作实践经验优先；

English gloss: Case 130: 5. Be familiar with basic Linux operations and principles; practical work experience in areas such as cloud computing and deep learning is preferred.

EX090 Current answer for case 130

[Return to the corresponding main rule](#)

Handbook example and treatment

Annotate Linux 基本操作 (basic Linux operations), S; 原理 (principles), K; 云计算 (cloud computing), S; and 深度学习 (deep learning), S. The last two types follow the local practical-work-experience requirement, which remains in context. The current answer uses four independent spans.

EX091 Case 134

[Return to the corresponding main rule](#)

Handbook example and treatment

案例 134 原句：任职要求：1、熟练掌握自然语言处理 NLP 相关理论与技术方法；

English gloss: Case 134: Job requirements: 1. Have a strong command of theories and technical methods related to natural language processing (NLP).

EX092 Current answer for case 134

[Return to the corresponding main rule](#)

Handbook example and treatment

The confirmed answer separates 自然语言处理 NLP 相关理论 (theories related to natural language processing/NLP), K, from 技术方法 (technical methods), K. The two items are joined by 与 (and), and NLP also qualifies 技术方法. The sentence describes theoretical and methodological knowledge. Do not create NLP 技术方法, which does not occur as a continuous substring. The earlier single K span was replaced by two K spans.

R15 Knowledge methods tools and distinct requirements within a sentence

EX093 Theory methods and testing practice

[Return to the corresponding main rule](#)

Handbook example and treatment

Theory/method contrast: in case 128, 软件工程理论和方法 (software-engineering theory and methods) is one general-knowledge K. The later clause explicitly concerns testing practice with white-/black-box examples: 软件测试的技术 (software-testing techniques), S; the second 理论 (theory), K; the second 方法 (methods), S; 白盒测试 (white-box testing), S; 黑盒测试 (black-box testing), S. Case 134 lacks this execution evidence, so 技术方法 (technical methods) is K. Linux 内核调试方法 (Linux kernel debugging methods) and 嵌入式软件设计

方法 (embedded-software design methods) directly name methods used to perform specific occupational tasks, S, rather than a generic methodological knowledge system.

Reading note: see cases 128 and 134 for full sentences and occurrence positions. The short method examples also appear in the correction supplement and R09.

EX094 Six tools in case 185

[Return to the corresponding main rule](#)

Handbook example and treatment

The tools in case 185 are qualified by both understanding principles and hands-on experience. The adopted answer is S; alternative K interpretations remain in review records. The qualifier applies to HDFS, Mangodb, Redis, Memcached, Kafka, and Dubbo. The preceding six knowledge items remain K. Preserve spelling.

Reading note: six specific tools labelled S do not make the preceding six general knowledge items S. See case 185 for the complete list.

EX095 Tool categories and stream computing in case 102

[Return to the corresponding main rule](#)

Handbook example and treatment

Assess technical categories case by case. In case 102, jvm, 关系型数据库 (relational databases), Nosql, RPC 框架 (RPC frameworks), 消息队列框架 (message-queue frameworks), and Zookeeper are S in the occupational tool-stack context. 流计算 (stream computing) is K as general technical knowledge. This is the project's conservative interpretation for this case. Separate annotation of a category depends on the requirement it expresses.

Reading note: Memcached is K in case 102 but S in case 185 because local contexts differ, not because a global term dictionary is inconsistent.

EX096 Local scope of shared practice in case 130

[Return to the corresponding main rule](#)

Handbook example and treatment

5. A qualifier may apply to nearby items joined by and/or, but not automatically to the whole advertisement. In case 130, practical work experience qualifies both cloud computing and deep learning. The shared experience wording may remain outside the entities while still guiding their interpretation. Explicit industry background follows the project rule for broad K; generic experience is excluded. Remove experience from complete activity names, but retain it when needed to preserve meaning under R08. Apply these conditions separately.

EX097 Algorithm fundamentals and locally elliptical principles

[Return to the corresponding main rule](#)

Handbook example and treatment

6. Complete knowledge phrases and local ellipsis. Retain explicit phrases such as 算法基础 (algorithm fundamentals). In case 130, 原理 (principles) refers locally back to Linux. Do not create discontinuous Linux 原理 or combine operations and principles merely to force one type.

R16 Willingness dispositions and research ability

EX098 Research willingness versus ordinary triggers

[Return to the corresponding main rule](#)

Handbook example and treatment

Case 160 retains 有意愿在 nlp 相关领域深入研究 (be willing to pursue in-depth research in NLP-related areas) as one T, expressing learning/development willingness. Do not remove willingness and recast 深入研究 (in-depth research) as S. See case 160 for the full source.

Reading note: retain 有意愿 (be willing), which expresses motivation; removing it and assigning S would change the meaning. In a general expression such as 有服务意识 (have a service-oriented mindset), still exclude 有 (have).

EX099 Product preferences and unresolved identity cases

[Return to the corresponding main rule](#)

Handbook example and treatment

Explicit job-related learning/development motivation differs from mere product preference. Do not add an entity for 喜欢高科技电子产品 (like high-tech electronic products) in case 169. This does not settle historical search-engine enthusiast or advanced-user identity cases automatically.

Reading note: search-engine enthusiasts and advanced users are mentioned only as historical identity cases with unconfirmed labels; no new label is assigned here.

EX100 Commercial sensitivity and data-oriented thinking

[Return to the corresponding main rule](#)

Handbook example and treatment

In case 195, 对商业和业务逻辑敏感 (sensitivity to commercial and business logic), 数据思维 (data-oriented thinking), and 数据决策意识 (awareness of data-driven decision-making) are T, as is 数据敏感度 (a feel for data) in case 197's disposition list. A domain word does not automatically turn general thinking or awareness into S. Specific data-analysis or modelling skills are assessed separately as S.

EX101 Academic research and unspecified competition experience

[Return to the corresponding main rule](#)

Handbook example and treatment

Case 158 retains the historical T for 学术研究 (academic research), which used the earlier research criteria. Do not reaffirm it merely because paper, experiment, or research project is absent. Before using the case under revised rules, check whether the research-experience wording supports a specific activity and record the confirmation. Do not infer a particular competition from unspecified 竞赛经验 (competition experience).

Reading note: apply the current handbook rules to academic research; an NLP job context does not justify inventing a specific competition when none is named.

R17 Occupational functions and general coordination

EX102 Project management and general management

[Return to the corresponding main rule](#)

Handbook example and treatment

Use across industries does not imply T. Project management in case 152 and product-development/translation-project contexts names an occupational function, S. General organizational management ability and leading a team are T. Project-management courses/principles may be K. [ESCO reusability levels](#) support the conceptual distinction but do not prescribe these Chinese labels.

Reading note: project management is not automatically T because it is used across industries. Course/principle contexts may support K.

EX103 Resource liaison and resource development

[Return to the corresponding main rule](#)

Handbook example and treatment

Classify resources and interfaces by the actual activity. In case 153, 翻译项目资源接口 (resource liaison for translation projects) denotes a specific resource-coordination duty within translation projects and is S. In case 154, 开发储备适用资源 (develop and build a pool of suitable resources) is also a specific occupational activity, S. 协调外部资源 (coordinate external resources) is T when it expresses general coordination. A domain name alone does not establish S.

EX104 Occupational quality assurance versus achieving targets

[Return to the corresponding main rule](#)

Handbook example and treatment

Case 124's original table separately labels 深度学习平台的测试 (testing of a deep-learning platform), S, and 质量保证 (quality assurance), S. These name testing and quality-assurance work, not a mere outcome promise. Do not mechanically remove 保证 (assurance) from the occupational name. See case 124 for the complete source.

EX105 General reflection and technical problem analysis

[Return to the corresponding main rule](#)

Handbook example and treatment

For words such as 分析 (analyze), identify the object from the current sentence or an adjacent clause within it. In case 117, 分析 follows 云产品现网疑难问题 (difficult issues in live cloud-product networks), so it denotes technical work and is S. In case 127, general reflection on work and proposed improvements are T. A technical job alone does not make every act of thinking, reflection, or communication S. In case 127, 质量意识 (quality awareness) and 测试理念 (a testing mindset) are separate T spans expressing awareness and attitudes; the specific work of developing test plans and testing strategies is S. See case 127 for exact boundaries.

EX106 Necessary objects and separable operations

[Return to the corresponding main rule](#)

Handbook example and treatment

Retain necessary objects. The three long technical activities in case 102 and 技术文档的编制及审核 (preparing and reviewing technical documentation) in case 179 follow the complete-case tables. Do not automatically extract preceding requirements-analysis, high-level-design, and detailed-design stages as independent duties. Case 103 retains 产品的探索 (product exploration), S, rather than 探索 alone. Case 120 separately displays 架构设计 (architecture design) and 运维 (operations and maintenance).

Reading note: this paragraph references several numbered cases, expanded separately. Case 120 permits splitting on the basis of complete meaning.

EX107 Shared heads in testing consulting and design

[Return to the corresponding main rule](#)

Handbook example and treatment

Do not lose necessary shared key words. Retain the shared wording that expresses the occupational activity in case 129's specialist-testing list, case 143's professional-consulting stages, and case 167's design list. Do not detach their objects and misread them as independent knowledge requirements, or insert absent action words.

R18 Review records and data versions

EX108 Differences in historical review methods

[Return to the corresponding main rule](#)

Handbook example and treatment

These cases were confirmed in different ways and must not all be understood as independent blind annotations. See R18 for the review-method distinctions and Appendix B for the historical 38 records and counts.

Reading note: confirmation methods and version changes follow R18. Use the cases according to their original review provenance.

R19 Separate empty and difficult records

EX109 Scope of confirmed-empty annotation

[Return to the corresponding main rule](#)

Handbook example and treatment

Handbook decision: read the source before deciding that a record is empty. Confirmed empty is the result of checking; record confirmed_empty with final labels []. A model output of [] means only that it returned no entities and may reflect omissions. R19 illustrates the check with 本科及以上学历 (a bachelor's degree or higher).

EX110 States of difficult records

[Return to the corresponding main rule](#)

Handbook example and treatment

Use `difficult_or_undetermined` for uncertain scope, boundaries, or types, missing characters, garbled text, or missing essential context. In review records, final labels is null and retained candidates are stored separately in `candidate_labels_preserved`. If legacy compatibility requires [] or partial candidates, also preserve the unresolved status and isolate the whole record. Do not treat an undecided answer as a confirmed negative example, or allow a trainer to convert null automatically into [].

Reading note: missing characters, garbled text, missing context, and absent entities are different states. Knowing some labels does not make an unresolved record automatically eligible for training.

EX111 Three confirmed-empty records

[Return to the corresponding main rule](#)

Handbook example and treatment

Historical records containing 功能解决对应问题; (the function solves the corresponding problem) or only 2. 其它职责同上。(2. Other duties as above) are empty under sentence-level extraction. This does not imply that preceding text has no skills. See R19 for full record IDs and restrictions.

Reading note: the two identical texts have different IDs and are both retained. Do not reconstruct absent entities from as above.

EX112 Missing activities or characters

[Return to the corresponding main rule](#)

Handbook example and treatment

Conversely, 基于分布式及大数据平台的优先考虑。(those based on distributed and big-data platforms are preferred) contains possible technical mentions but lacks the essential activity; do not treat it as a pure cross-reference and mark it empty. 有? (have?) and garbled text cannot be reliably interpreted and are unresolved. These are project examples, not ESCO rules for Chinese boundaries.

Reading note: neither example is confirmed empty. The handbook supplies no complete usable labels or offsets here.

R20 Uses of model review and human confirmation

EX113 Scope of model review

[Return to the corresponding main rule](#)

Handbook example and treatment

The historical 53 model-resolved records and three groups of identical-text conflicts are preserved in the model case log, with confirmation method recorded as model review. Reassess the same terms in new contexts. Save later human changes as a new version and retain the previous one.

Reading note: the companion log describes each of the 53 model-reviewed cases. This entry explains their confirmation method and permitted use.

R21 Scope evidence and preservation of local meaning

EX114 Unclear company description versus job requirement

[Return to the corresponding main rule](#)

Handbook example and treatment

Use only the current sentence in this version, including its component clauses. Do not consult preceding or following sentences, the job title, or the full advertisement to supply missing meaning. Record IDs and queue metadata serve only to locate records. If the sentence does not establish annotation scope, request review. Full historical advertisements remain available for provenance, not as additional context for current decisions.

EX115 Understanding business customer needs and product-line responsibility

[Return to the corresponding main rule](#)

Handbook example and treatment

Distinguish introductory triggers from actual action heads. Usually exclude familiar with or possess, but do not remove the only expression of occupational activity or experience and leave an object that cannot express a competence. For boundaries involving understanding business, customer needs, or responsibility for a product line, examine the current duty and explain why an action is necessary. This does not make every understand S or include every responsible for in a span.

Reading note: these are boundary questions, not unconditional labels for 理解业务 (understand the business), 客户需求 (customer needs), or 负责产品线 (be responsible for a product line).

EX116 Object lists following such as or including

[Return to the corresponding main rule](#)

Handbook example and treatment

Apply R21 to determine whether category terms and specifics each express a requirement. Do not add an entity for a category that only introduces a list. Independent requirements may be separate, but final spans must not overlap.

Reading note: conceptual inclusion and overlapping character spans are different issues. Each item's independent requirement status still depends on context.

R23 Splitting coordinated items under shared experience context

EX117 Coordination under shared experience context

[Return to the corresponding main rule](#)

Handbook example and treatment

When items joined by 或 (or), 和 (and), or 与 (and) name independent activities, tools, or other objects of practical work, check whether a shared relevant-work-experience phrase in the same sentence applies to all of them. If it does and splitting preserves their meaning, annotate separate continuous source spans. The shared experience phrase may remain outside the spans while guiding their interpretation. Classify each item using the practical-experience requirement in the sentence.

EX118 Experience and BI boundaries

[Return to the corresponding main rule](#)

Handbook example and treatment

Under R21, first identify which coordinated items the shared phrase qualifies, then test whether splitting preserves the requirement. Separate independently meaningful items; retain necessary actions, objects, or experience wording. Record alternatives and request review if uncertainty remains.

Reading note: check the experience phrase's scope, post-split completeness, and type together.

EX119 Source sentence of case 194

[Return to the corresponding main rule](#)

Handbook example and treatment

4、三年以上大中型互联网公司经营或 BI 相关工作经验，互联网公司广告运营分析优先；

English gloss: 4. At least three years of relevant work experience in internet company operations or BI at medium-sized or large companies; experience in advertising operations analysis at internet companies preferred.

EX120 First new span in case 194

[Return to the corresponding main rule](#)

Handbook example and treatment

[9,16) → 互联网公司经营 → S

English gloss: [9,16) → internet company operations → S.

EX121 Second new span in case 194

[Return to the corresponding main rule](#)

Handbook example and treatment

[17,19) → BI → S

EX122 Third span in case 194

[Return to the corresponding main rule](#)

Handbook example and treatment

[26,37) → 互联网公司广告运营分析 → S (此项不变)

English gloss: [26,37) → advertising operations analysis at internet companies → S (unchanged).

EX123 Choosing boundaries for BI and shared experience

[Return to the corresponding main rule](#)

Handbook example and treatment

This example selects BI, S, leaving shared 相关工作经验 (relevant work experience) in context. BI 相关工作经验 is continuous source text at [17,25), but is not the chosen boundary. See case 194 for the three current spans and full source.

Reading note: adding words absent from the source differs from excluding source words from a selected entity.

EX124 Earlier annotation for case 194

[Return to the corresponding main rule](#)

Handbook example and treatment

This entry shows the old A100 long span [9,25). The three S spans in R23 are later guidance. See case 194 for the full comparison; do not write later guidance back into historical data.

Reading note: this row is a historical answer only. Current guidance uses the three R23 spans; historical production data remain unchanged by this compilation.

Part II Full source texts and answers for numbered cases

Cases follow their first citation in the main text. Case numbers identify the original records. Use them to learn boundary and type decisions. Disputed entries retain historical answers with nearby questions for discussion. Character intervals start at zero and exclude the right endpoint.

Source text colour key: **L** Language proficiency **K** Knowledge **S** Occupational skill **T** Transversal competence

Highlighted and underlined Chinese text marks selected spans. Matching colours identify L, K, S, and T; the Type column records the label explicitly. English glosses are explanatory, not extraction targets. Character intervals always refer to the unchanged Chinese source.

Case number	Original record ID	First linked entry
130	1825-s0008	EX089
134	1827-s0004	EX091
128	1825-s0006	EX093
185	1836-s0012	EX094
102	1821-s0000	EX095
160	1833-s0004	EX098
169	1834-s0010	EX099
195	1838-s0009	EX100
197	1838-s0012	EX100
158	1833-s0002	EX101
152	1831-s0010	EX102
153	1832-s0001	EX103
154	1832-s0002	EX103
124	1825-s0000	EX104
127	1825-s0005	EX105
117	1824-s0000	EX105
179	1836-s0004	EX106
103	1822-s0000	EX106
120	1824-s0005	EX106

Case number	Original record ID	First linked entry
129	1825-s0007	EX107
143	1829-s0002	EX107
167	1834-s0008	EX107
194	1838-s0008	EX119

Case 130 Linux operations principles and shared practical experience

Record ID: 1825-s0008. Related entry: [EX089](#).

Full Chinese source

5.熟悉 **Linux 基本操作**和**原理**，在**云计算**、**深度学习**等领域者有工作实践经验优先；

English gloss: 5. Be familiar with basic Linux operations and principles; practical work experience in areas such as cloud computing and deep learning is preferred.

Chinese source span and English gloss	Type	Chinese character interval
Linux 基本操作 <i>English gloss: Basic Linux operations</i>	S	[4,13)
原理 <i>English gloss: Principles</i>	K	[14,16)
云计算 <i>English gloss: Cloud computing</i>	S	[18,21)
深度学习 <i>English gloss: Deep learning</i>	S	[22,26)

Rationale: the split boundaries are confirmed. Linux 基本操作 (basic Linux operations) is S and 原理 (principles) is K. 云计算 (cloud computing) and 深度学习 (deep learning) are separate S spans under the explicit practical-work-experience requirement, not fixed domain-name labels.

Case 134 NLP theory and general technical methods

Record ID: 1827-s0004. Related entry: [EX091](#).

Full Chinese source

任职要求： 1、熟练掌握**自然语言处理 NLP 相关理论**与**技术方法**；

English gloss: Job requirements: 1. Have a strong command of theories and technical methods related to natural language processing (NLP).

Chinese source span and English gloss	Type	Chinese character interval
<u>自然语言处理 NLP 相关理论</u> <i>English gloss: Theories related to natural language processing (NLP)</i>	K	[13,26)
<u>技术方法</u> <i>English gloss: Technical methods</i>	K	[27,31)

Rationale: the confirmed split yields 自然语言处理 NLP 相关理论 (NLP-related theories), K, and 技术方法 (technical methods), K. The sentence describes a general knowledge system without explicit execution or practice.

Case 128 Software-engineering theory and testing practice

Record ID: 1825-s0006. Related entry: [EX093](#).

Full Chinese source

4. 熟悉软件工程理论和方法，对各种软件测试的技术、理论、方法都有实践经验，如白盒测试、黑盒测试。

English gloss: 4. Be familiar with software-engineering theory and methods, and have practical experience with various software-testing techniques, theories, and methods, such as white-box and black-box testing.

Chinese source span and English gloss	Type	Chinese character interval
<u>软件工程理论和方法</u> <i>English gloss: Software-engineering theory and methods</i>	K	[4,13)
<u>软件测试的技术</u> <i>English gloss: Software-testing techniques</i>	S	[17,24)
<u>理论</u> <i>English gloss: Theory</i>	K	[25,27)
<u>方法</u> <i>English gloss: Methods</i>	S	[28,30)
<u>白盒测试</u> <i>English gloss: White-box testing</i>	S	[38,42)
<u>黑盒测试</u> <i>English gloss: Black-box testing</i>	S	[43,47)

Rationale: software-engineering theory and methods form one K. Later testing techniques and methods have practice evidence and specific examples, S. Explicit theory is K; white-box and black-box testing are S.

Case 185 Tools qualified by knowledge and practice

Record ID: 1836-s0012. Related entry: [EX094](#).

Full Chinese source

5. 熟悉网络应用架构和系统架构，对分布式缓存、消息队列、大数据处理、远程调用等技术至少有 2 项有研究，熟悉 HDFS、Mangodb、Redis、Memcached、Kafka、Dubbo 中的 2 项以上的技术，了解其原理和使用其优缺点，并有实战经验者优先；

English gloss: 5. Be familiar with network-application and system architectures; have studied at least two technologies such as distributed caching, message queues, big-data processing, or remote calls; be familiar with at least two of HDFS, Mangodb, Redis, Memcached, Kafka, and Dubbo; understand their principles and the advantages and disadvantages of using them. Hands-on experience is preferred.

Chinese source span and English gloss	Type	Chinese character interval
<u>网络应用架构</u> English gloss: Network-application architecture	K	[6,12)
<u>系统架构</u> English gloss: System architecture	K	[13,17)
<u>分布式缓存</u> English gloss: Distributed caching	K	[19,24)
<u>消息队列</u> English gloss: Message queues	K	[25,29)
<u>大数据处理</u> English gloss: Big-data processing	K	[30,35)
<u>远程调用</u> English gloss: Remote calls	K	[36,40)
<u>HDFS</u>	S	[54,58)
<u>Mangodb</u>	S	[59,66)
<u>Redis</u>	S	[67,72)
<u>Memcached</u>	S	[73,82)
<u>Kafka</u>	S	[84,89)
<u>Dubbo</u>	S	[90,95)

Rationale: the six specific tools are qualified by understanding principles and explicit hands-on experience; the adopted type is S. Knowledge interpretations remain in review records. The preceding six general knowledge items remain K.

Contrast: distinguish the preceding knowledge content from the later tool-practice requirement. Preserve the original spelling Mangodb.

Case 102 Development-role tool stacks knowledge and necessary long spans

Record ID: 1821-s0000. Related entry: [EX095](#).

Full Chinese source

工作职责: 1.负责 AI 项目的系统设计和开发工作,和算法工程师、研发、交付等合作,将 AI 能力应用到实际客户应用中 2.深入理解公司的 AI 平台及产品框架,并对平台进行改进、扩展,以服务更多的应用场景 3.核心代码的编写、测试和评审 4.参与解决方案或产品的需求设计,深入理解业务、参与需求分析并将之转换成技术实现方案 5.攻克高并发、高稳定性,业务模型复杂等带来的各种挑战及技术难关 6.研究相关新技术,并努力提升项目研发的效率和质量 任职要求: 1.计算机相关专业,本科及以上学历 2.5 年以上的产品研发或大型项目工程开发经验 3.精通 Java 开发语言,熟悉 jvm, web 开发、缓存、分布式、高并发等核心技术 4.熟悉 Spring 框架,熟练掌握 SpringCloud、SpringMVC、Hibernate 和 MyBatis 等 5.扎实的算法、数据结构功底以及系统设计能力,有丰富的性能调优经验 6.熟悉 Docker/Kubernetes 相关生态并有实践经验,有集成 Openstack 和 Docker 落地经验者优先 7.熟练掌握常见的关系型数据库(如 Mysql)和 Nosql(如 Redis)数据库,熟悉常见的缓存技术与实现原理(如 Memcached 等) 8.熟悉 RPC 框架(如 Dubbo、Thrift 等)、消息队列框架(如 Kafka、RocketMQ 等),熟悉 Hadoop/Spark,流计算等,了解分布式协调服务 Zookeeper 9.熟悉权限管理、统一登录等系统的设计和实现原理,了解 Oauth 协议,有开放平台设计和开发经验者优先 10.熟悉 Linux 系统原理、熟练运用 Linux 命令与脚本 11.理解敏捷开发流程并有实践经验 12.为人乐观积极,具有良好的沟通和团队协作能力 福利待遇 年底双薪,股票期权,带薪年假,交通补助,通讯津贴,午餐补助,绩效奖金,补充公积金

English gloss: Duties: 1. Design and develop AI-project systems; cooperate with algorithm engineers, R&D, and delivery staff; apply AI capabilities to real client applications. 2. Understand the company's AI platform and product framework in depth; improve and extend the platform for more application scenarios. 3. Write, test, and review core code. 4. Participate in requirements design for solutions or products, understand the business, analyze requirements, and turn them into technical implementation plans. 5. Overcome challenges and technical difficulties arising from high concurrency, high stability requirements, and complex business models. 6. Research new technologies and improve project R&D efficiency and quality. Requirements: 1. A bachelor's degree or higher, with a major in computer science or a related field. 2. Five years or more of product R&D or large-project engineering development experience. 3. Proficiency in Java development; familiarity with jvm and core technologies including web development, caching, distributed systems, and high concurrency. 4. Familiarity with Spring and proficiency in SpringCloud, SpringMVC, Hibernate, MyBatis, etc. 5. Solid algorithms and data-structure foundations, system-design ability, and extensive performance-tuning experience. 6. Familiarity and practical experience with the Docker/Kubernetes ecosystem; experience implementing Openstack and Docker integration is preferred. 7. Proficiency with common relational databases such as Mysql and Nosql databases such as Redis; familiarity with caching technologies and implementation principles such as Memcached. 8. Familiarity with RPC frameworks such as Dubbo and Thrift, message-queue frameworks such as Kafka and RocketMQ, Hadoop/Spark and stream computing; knowledge of the distributed coordination service Zookeeper. 9. Familiarity with the design and implementation principles of permission-management and single-sign-on systems; knowledge of Oauth; open-platform design and development experience is preferred. 10. Familiarity with Linux system principles and proficient use of Linux commands and scripts. 11. Understanding and practical experience of agile development processes. 12. An optimistic, proactive disposition, good communication, and teamwork ability. Benefits: year-end double salary, stock options, paid annual leave, transport and communications allowances, lunch allowance, performance bonuses, and supplementary housing-fund contributions.

Chinese source span and English gloss	Type	Chinese character interval
AI 项目的系统设计和开发 English gloss: AI-project system design and development	S	[9,21)

Chinese source span and English gloss	Type	Chinese character interval
<u>合作</u> English gloss: Cooperation	T	[37,39)
<u>将 AI 能力应用到实际客户应用中</u> English gloss: Applying AI capabilities to real client applications	S	[40,55)
<u>AI 平台</u> English gloss: AI platform	K	[64,68)
<u>产品框架</u> English gloss: Product framework	K	[69,73)
<u>改进</u> English gloss: Improvement	S	[80,82)
<u>扩展</u> English gloss: Extension	S	[83,85)
<u>核心代码的编写</u> English gloss: Writing core code	S	[98,105)
<u>测试</u> English gloss: Testing	S	[106,108)
<u>评审</u> English gloss: Review	S	[109,111)
<u>解决方案或产品的需求设计</u> English gloss: Requirements design for solutions or products	S	[115,127)
<u>需求分析</u> English gloss: Requirements analysis	S	[137,141)
<u>将之转换成技术实现方案</u> English gloss: Turning them into technical implementation plans	S	[142,153)
<u>攻克高并发、高稳定性，业务模型复杂等带来的各种挑战及技术难关</u> English gloss: Overcoming challenges and technical difficulties arising from high concurrency, high stability requirements, and complex business models	S	[155,185)
<u>研究相关新技术</u> English gloss: Researching relevant new technologies	S	[187,194)

Chinese source span and English gloss	Type	Chinese character interval
<u>计算机相关专业</u> English gloss: A major in computer science or a related field	K	[217,224]
<u>本科及以上学历</u> English gloss: Bachelor's degree or higher	K	[225,232]
<u>产品研发</u> English gloss: Product research and development	S	[239,243]
<u>大型项目工程开发</u> English gloss: Large-project engineering development	S	[244,252]
<u>Java</u>	S	[258,262]
<u>jvm</u>	S	[269,272]
<u>web 开发</u> English gloss: Web development	S	[273,278]
<u>缓存</u> English gloss: Caching	K	[279,281]
<u>分布式</u> English gloss: Distributed systems	K	[282,285]
<u>高并发</u> English gloss: High concurrency	K	[286,289]
<u>Spring</u>	S	[298,304]
<u>SpirngCloud</u>	S	[311,322]
<u>SpringMVC</u>	S	[323,332]
<u>Hibernate</u>	S	[333,342]
<u>MyBatis</u>	S	[343,350]
<u>算法</u> English gloss: Algorithms	K	[356,358]
<u>数据结构功底</u> English gloss: Foundation in data structures	K	[359,365]
<u>系统设计能力</u> English gloss: System-design ability	S	[367,373]

Chinese source span and English gloss	Type	Chinese character interval
<u>性能调优</u> <i>English gloss: Performance tuning</i>	S	[378,382)
<u>Docker</u>	S	[388,394)
<u>Kubernetes</u>	S	[395,405)
<u>集成 Openstack 和 Docker 落地</u> <i>English gloss: Implementing Openstack and Docker integration</i>	S	[417,437)
<u>关系型数据库</u> <i>English gloss: Relational databases</i>	S	[451,457)
<u>Mysql</u>	S	[459,464)
<u>Nosql</u>	S	[466,471)
<u>Redis</u>	S	[473,478)
<u>缓存技术与实现原理</u> <i>English gloss: Caching technologies and implementation principles</i>	K	[488,497)
<u>Memcached</u>	K	[499,508)
<u>RPC 框架</u> <i>English gloss: RPC frameworks</i>	S	[514,519)
<u>Dubbo</u>	S	[521,526)
<u>Thrift</u>	S	[527,533)
<u>消息队列框架</u> <i>English gloss: Message-queue frameworks</i>	S	[536,542)
<u>Kafka</u>	S	[544,549)
<u>RocketMQ</u>	S	[550,558)
<u>Hadoop</u>	S	[563,569)
<u>Spark</u>	S	[570,575)
<u>流计算</u> <i>English gloss: Stream computing</i>	K	[576,579)
<u>Zookeeper</u>	S	[590,599)

Chinese source span and English gloss	Type	Chinese character interval
<u>权限管理、统一登录等系统的设计和实现原理</u> <i>English gloss: Design and implementation principles of permission-management, single-sign-on, and related systems</i>	K	[603,623)
<u>Oauth 协议</u> <i>English gloss: Oauth protocol</i>	K	[626,633)
<u>开放平台设计和开发</u> <i>English gloss: Open-platform design and development</i>	S	[635,644)
<u>Linux 系统原理</u> <i>English gloss: Linux system principles</i>	K	[654,663)
<u>Linux 命令</u> <i>English gloss: Linux commands</i>	S	[668,675)
<u>脚本</u> <i>English gloss: Scripts</i>	S	[676,678)
<u>敏捷开发流程</u> <i>English gloss: Agile development process</i>	S	[683,689)
<u>乐观积极</u> <i>English gloss: Optimistic and proactive</i>	T	[700,704)
<u>沟通</u> <i>English gloss: Communication</i>	T	[710,712)
<u>团队协作能力</u> <i>English gloss: Teamwork ability</i>	T	[713,719)

Rationale: tool-stack items retain S in the occupational-use context; 流计算 (stream computing) is K as a general technical-knowledge category. This is the project's conservative interpretation of this case. Three necessary long technical-activity spans retain S. 深入理解业务 (understand the business in depth) remains unannotated as a separate entity.

Contrast: Memcached is K here and S in case 185 because local requirements differ. Preserve source spellings such as SpringCloud and Nosql.

Case 160 Complete research-willingness expression

Record ID: 1833-s0004. Related entry: [EX098](#).

Full Chinese source

3. 性格开朗，乐于分享和交流，有意愿在 nlp 相关领域深入研究。

English gloss: 3. Have an outgoing personality, enjoy sharing and communicating, and be willing to pursue in-depth research in NLP-related areas.

Chinese source span and English gloss	Type	Chinese character interval
<u>性格开朗</u> <i>English gloss: Outgoing personality</i>	T	[3,7)
<u>乐于分享和交流</u> <i>English gloss: Willingness to share and communicate</i>	T	[8,15)
<u>有意愿在 nlp 相关领域深入研究</u> <i>English gloss: Willingness to pursue in-depth research in NLP-related areas</i>	T	[16,31)

Rationale: retain the complete research-willingness expression as T; the other two T spans remain unchanged. This decision draws on ESCO's willingness-to-learn concept and uses the conditional-confirmation method recorded for the case.

Case 169 Product preferences and occupational dispositions

Record ID: 1834-s0010. Related entry: [EX099](#).

Full Chinese source

5、细心、工作负责、团队协作精神强，有热情，喜欢高科技电子产品；

English gloss: 5. Be attentive, responsible at work, and strongly team-oriented; show enthusiasm and like high-tech electronic products.

Chinese source span and English gloss	Type	Chinese character interval
<u>细心</u> <i>English gloss: Attentiveness</i>	T	[2,4)
<u>工作负责</u> <i>English gloss: Responsibility at work</i>	T	[5,9)
<u>团队协作精神</u> <i>English gloss: A collaborative attitude</i>	T	[10,16)
<u>热情</u> <i>English gloss: Enthusiasm</i>	T	[19,21)

Rationale: retain the four existing T spans. Liking high-tech electronic products expresses only product preference, without learning or occupational action, so no knowledge or skill is added. This differs from case 160's willingness to undertake research.

Case 195 Commercial and data-related thinking dispositions

Record ID: 1838-s0009. Related entry: [EX100](#).

Full Chinese source

5、对商业和业务逻辑敏感，具备良好的逻辑分析能力和系统性思维能力，优秀的数据思维和数据决策意识，且思维缜密，做事认真严谨；

English gloss: 5. Be sensitive to commercial and business logic; have good logical-analysis and systematic-thinking abilities, strong data-oriented thinking and awareness of data-driven decisions; think carefully and work conscientiously and rigorously.

Chinese source span and English gloss	Type	Chinese character interval
<u>对商业和业务逻辑敏感</u> English gloss: Sensitivity to commercial and business logic	T	[2,12)
<u>逻辑分析能力</u> English gloss: Logical-analysis ability	T	[18,24)
<u>系统性思维能力</u> English gloss: Systematic-thinking ability	T	[25,32)
<u>数据思维</u> English gloss: Data-oriented thinking	T	[36,40)
<u>数据决策意识</u> English gloss: Awareness of data-driven decision-making	T	[41,47)
<u>思维缜密</u> English gloss: Careful and thorough thinking	T	[49,53)
<u>做事认真严谨</u> English gloss: Conscientious and rigorous work	T	[54,60)

Rationale: commercial/business-logic sensitivity, data-oriented thinking, and data-driven-decision awareness are T. The sentence emphasizes sensitivity, ways of thinking, and awareness. Assess specific techniques such as data modelling or analysis separately in their occupational contexts.

Case 197 Advertising optimization and a feel for data

Record ID: 1838-s0012. Related entry: [EX100](#).

Full Chinese source

一句话介绍工作内容: base 在广告销售团队, 协助销售, 指导客户广告投放, 优化广告效果, 提升广告主的消耗以完成团队的销售目标
 特质: 踏实、细心、有数据敏感度、抗压能力强

English gloss: The role in one sentence: based in the advertising sales team, support sales, advise clients on ad placement, optimize advertising performance, and increase advertisers' spending to meet the team's sales targets.
 Personal qualities: dependable and attentive to detail, with a feel for data and the ability to work under pressure.

Chinese source span and English gloss	Type	Chinese character interval
<u>协助销售</u> English gloss: Sales support	S	[22,26)
<u>指导客户广告投放</u> English gloss: Advising clients on ad placement	S	[27,35)

Chinese source span and English gloss	Type	Chinese character interval
<u>优化广告效果</u> <i>English gloss: Optimizing advertising performance</i>	S	[36,42)
<u>踏实</u> <i>English gloss: Dependability</i>	T	[64,66)
<u>细心</u> <i>English gloss: Attentiveness</i>	T	[67,69)
<u>数据敏感度</u> <i>English gloss: A feel for data</i>	T	[71,76)
<u>抗压能力</u> <i>English gloss: Ability to work under pressure</i>	T	[77,81)

Rationale: 协助销售 (support sales), 指导客户广告投放 (advise clients on ad placement), and 优化广告效果 (optimize advertising performance) each describe occupational work and are S. Although the activities may stand in a general-to-specific relationship, each has an identifiable meaning and the source spans do not overlap, so all three are retained. 提升广告主的消耗以完成团队的销售目标 (increase advertisers' spending to meet the team's sales targets) states a performance objective and is not separately annotated. 踏实 (being dependable and conscientious), 细心 (attention to detail), 数据敏感度 (a feel for data), and 抗压能力 (ability to work under pressure) express personal qualities or general abilities and are T. The decision turns on specific work versus a result target, not a fixed classification of verbs such as optimize or increase.

Case 158 Academic research and unspecified competition experience

Record ID: 1833-s0002. Related entry: [EX101](#).

Full Chinese source

要求: 1.熟悉 nlp 领域基本知识, 了解 nlp 领域某一方面先进技术, 对分类、匹配、生成模型、强化学习等有深入理解优先, 具备相关学术研究经验或竞赛经验优先。

English gloss: Requirements: 1. Be familiar with basic NLP knowledge and understand advanced technology in some area of NLP. In-depth understanding of classification, matching, generative models, reinforcement learning, etc. is preferred, as is relevant academic-research or competition experience.

Chinese source span and English gloss	Type	Chinese character interval
<u>nlp 领域基本知识</u> <i>English gloss: Basic knowledge of NLP</i>	K	[8,17)
<u>nlp 领域某一方面先进技术</u> <i>English gloss: Advanced technology in a particular NLP area</i>	K	[20,33)
<u>分类</u> <i>English gloss: Classification</i>	K	[35,37)

Chinese source span and English gloss	Type	Chinese character interval
<u>匹配</u> <i>English gloss: Matching</i>	K	[38,40)
<u>生成模型</u> <i>English gloss: Generative models</i>	K	[41,45)
<u>强化学习</u> <i>English gloss: Reinforcement learning</i>	K	[46,50)
<u>学术研究</u> <i>English gloss: Academic research</i>	T	[63,67)

Historical annotation note: academic research was T under the research-ability conditions used at the time. The revised proposal no longer requires three fixed words, so this T is an illustration of the historical annotation, not a claim that the new rules necessarily yield T. Review the meaning of the research experience before reusing it under the new rules. The source names no competition type; do not add a specific competition.

Scope: this case illustrates the previous version. The answer under the revised rules for academic research still requires case-specific confirmation and is not a confirmed reference answer under the revised rules.

Case 152 Project management and general management ability

Record ID: 1831-s0010. Related entry: [EX102](#).

Full Chinese source

5、优秀的沟通协调及组织管理能力，具备目标设定与执行、项目管理、激励与辅导、有促进他人行动的能力，能够领导团队完成平台级别的开发任务。

English gloss: 5. Have excellent communication, coordination, and organizational-management abilities; be capable of goal setting and execution, project management, motivation and coaching, and prompting others to act; lead a team to complete platform-level development tasks.

Chinese source span and English gloss	Type	Chinese character interval
<u>沟通协调</u> <i>English gloss: Communication and coordination</i>	T	[5,9)
<u>组织管理能力</u> <i>English gloss: Organizational-management ability</i>	T	[10,16)
<u>目标设定与执行</u> <i>English gloss: Goal setting and execution</i>	T	[19,26)
<u>项目管理</u> <i>English gloss: Project management</i>	S	[27,31)

Chinese source span and English gloss	Type	Chinese character interval
<u>激励与辅导</u> <i>English gloss: Motivation and coaching</i>	T	[32,37)
<u>促进他人行动的能力</u> <i>English gloss: Ability to prompt others to act</i>	T	[39,48)
<u>领导团队</u> <i>English gloss: Leading a team</i>	T	[51,55)

Rationale: project management names an occupational management function, S; the other general management and leadership abilities are T. This decision draws on ESCO concepts and uses the conditional-confirmation method recorded for the case.

Case 153 Resource liaison for translation projects

Record ID: 1832-s0001. Related entry: [EX103](#).

Full Chinese source

- 2、翻译项目资源接口，参与项目管理；

English gloss: 2. Serve as the resource liaison for translation projects and participate in project management.

Chinese source span and English gloss	Type	Chinese character interval
<u>翻译项目资源接口</u> <i>English gloss: Resource liaison for translation projects</i>	S	[2,10)
<u>项目管理</u> <i>English gloss: Project management</i>	S	[13,17)

Confirmed decision: 翻译项目资源接口 (resource liaison for translation projects) denotes a specific resource-coordination duty within translation projects and is S; 项目管理 (project management) is S. The former had the historical label T. This revision changes the handbook's illustrative answer, not saved data. A translation-domain qualifier alone does not make other general communication activities S.

Case 154 Translation-project resource development

Record ID: 1832-s0002. Related entry: [EX103](#).

Full Chinese source

- 3、利用公司现有资源渠道，根据翻译项目需求开发储备适用资源；

English gloss: 3. Use the company's existing resource channels to develop and build a pool of suitable resources for translation-project needs.

Chinese source span and English gloss	Type	Chinese character interval
<u>开发储备适用资源</u> <i>English gloss: Developing and building a pool of suitable resources</i>	S	[21,29)

Rationale: developing and building suitable resources is S; translation-project needs explicitly qualify the resource-development activity. General personnel liaison is assessed separately as transversal coordination.

Case 124 Occupational quality assurance is not a pure outcome

Record ID: 1825-s0000. Related entry: [EX104](#).

Full Chinese source

工作职责：1.主要负责旷视深度学习平台的测试和质量保证工作。

English gloss: Duties: 1. Primarily responsible for testing and quality assurance of Megvii's deep-learning platform.

Chinese source span and English gloss	Type	Chinese character interval
<u>深度学习平台的测试</u> English gloss: Testing of a deep-learning platform	S	[13,22)
<u>质量保证</u> English gloss: Quality assurance	S	[23,27)

Rationale: testing and quality assurance in this sentence refer to occupational work on the platform. Label 深度学习平台的测试 (testing a deep-learning platform) and 质量保证 (quality assurance) separately as S. Apply the scope rule separately to outcome clauses that merely promise to meet a quality standard.

Case 127 Testing activities and general work reflection

Record ID: 1825-s0005. Related entry: [EX105](#).

Full Chinese source

3.强烈的质量意识和测试理念，能够根据项目特点、用户需求等制定完善的项目测试计划和测试方案，善于总结项目、团队工作中存在的问题，并能够提出有效的改进办法；

English gloss: 3. Have strong quality awareness and a testing mindset; develop comprehensive project test plans and testing strategies based on project characteristics, user needs, etc.; be good at reviewing problems in project and team work and proposing effective improvements.

Chinese source span and English gloss	Type	Chinese character interval
<u>质量意识</u> English gloss: Quality awareness	T	[5,9)
<u>测试理念</u> English gloss: A testing mindset	T	[10,14)
<u>制定完善的项目测试计划和测试方案</u> English gloss: Developing comprehensive project test plans and testing strategies	S	[29,45)
<u>总结项目、团队工作中存在的问题</u> English gloss: Reviewing problems in project and team work	T	[48,63)

Chinese source span and English gloss	Type	Chinese character interval
<u>提出有效的改进办法</u> <i>English gloss: Proposing effective improvements</i>	T	[67,76)

Rationale: 质量意识 (quality awareness) and 测试理念 (a testing mindset) express awareness and attitudes toward testing in this sentence and are separate T spans. 制定完善的项目测试计划和测试方案 (develop comprehensive project test plans and testing strategies) is specific testing work, S. 总结项目、团队工作中存在的问题 (review problems in project and team work) and 提出有效的改进办法 (propose effective improvements) express general reflection and improvement abilities, T. Assess each item by its meaning in this sentence, not by assuming that coordinated items share a type. [Project rule]

Case 117 Analysis in a technical-problem context

Record ID: 1824-s0000. Related entry: [EX105](#).

Full Chinese source

【工作职责】1.负责解决云产品现网疑难问题，深入分析，推动问题解决方案落地；

English gloss: [Duties] 1. Resolve difficult live-network problems involving cloud products, analyze them in depth, and advance the implementation of solutions.

Chinese source span and English gloss	Type	Chinese character interval
<u>解决云产品现网疑难问题</u> <i>English gloss: Resolving difficult live-network problems involving cloud products</i>	S	[10,21)
<u>分析</u> <i>English gloss: Analysis</i>	S	[24,26)
<u>推动问题解决方案落地</u> <i>English gloss: Advancing the implementation of problem solutions</i>	S	[27,37)

Rationale: the three independent activities are separate S spans. 分析 (analyze) refers back to the immediately preceding 云产品现网疑难问题 (difficult issues in live cloud-product networks), giving it a specific technical object.

Case 179 Technical documentation and development phases

Record ID: 1836-s0004. Related entry: [EX106](#).

Full Chinese source

5、参与需求分析、概要设计、详细设计等开发过程中相关技术文档的编制及审核；

English gloss: 5. Participate in preparing and reviewing technical documentation during requirements analysis, high-level design, detailed design, and other development phases.

Chinese source span and English gloss	Type	Chinese character interval
<u>技术文档的编制及审核</u> <i>English gloss: Preparation and review of technical documentation</i>	S	[27,37)

Rationale: retain 技术文档的编制及审核 (preparation and review of technical documentation), S. Requirements analysis, high-level design, and detailed design qualify the documentation's development phases; the sentence does not independently require those activities, so add no extra annotations for them.

Case 103 Product exploration and general coordination

Record ID: 1822-s0000. Related entry: [EX106](#).

Full Chinese source

岗位职责：1.参与设计下一代的人工智能开发和应用平台，负责结合人工智能和垂直行业场景的产品的探索、机会发现、商务洽谈与方案及产品设计，协调各方资源推动产品上线;

English gloss: Duties: 1. Participate in designing the next generation of AI development and application platforms. Undertake product exploration, opportunity discovery, business negotiation, and solution/product design combining AI with vertical-industry scenarios. Coordinate resources from different parties to advance product launch.

Chinese source span and English gloss	Type	Chinese character interval
设计下一代的人工智能开发和应用平台 English gloss: Designing next-generation AI development and application platforms	S	[9,26)
产品的探索 English gloss: Product exploration	S	[43,48)
机会发现 English gloss: Opportunity discovery	S	[49,53)
商务洽谈 English gloss: Business negotiation	T	[54,58)
方案及产品设计 English gloss: Solution and product design	S	[59,66)
协调各方资源 English gloss: Coordinating resources from different parties	T	[67,73)
推动产品上线 English gloss: Advancing product launch	S	[73,79)

Rationale: the source lists product exploration, identifying opportunities, solution and product design, and bringing products to launch. Historical review interpreted 推动产品上线 (bring products to launch) as a delivery responsibility and retained S. The word 上线 (launch) alone does not justify extending that decision to pure goal statements. 产品的探索 (product exploration) is S, retaining the occupational object 产品的 (of products). Solution and product design retain the shared head 设计 (design). Business negotiations and coordinating resources across parties remain T.

Case 120 Separating architecture design and maintenance

Record ID: 1824-s0005. Related entry: [EX106](#).

Full Chinese source

2.有云计算大规模基础服务平台的架构设计、运维经验优先；

English gloss: 2. Experience in architecture design and operations/maintenance of large-scale cloud-computing infrastructure service platforms is preferred.

Chinese source span and English gloss	Type	Chinese character interval
云计算大规模基础服务平台的架构设计 <i>English gloss: Architecture design for large-scale cloud-computing infrastructure service platforms</i>	S	[3,20)
运维 <i>English gloss: Operations and maintenance</i>	S	[21,23)

Rationale: architecture design and operations/maintenance are independent occupational activities. Separate 云计算大规模基础服务平台的架构设计 (architecture design for large-scale cloud-computing infrastructure service platforms), S, and 运维 (operations and maintenance), S. The latter expresses a skill without added words; exclude experience.

Boundary note: both activities independently express requirements here. Compare cases 129, 143, and 167 to test whether splitting loses necessary meaning.

Case 129 Retaining a shared head in specialist testing

Record ID: 1825-s0007. Related entry: [EX107](#).

Full Chinese source

在自动化、服务器性能、安全等专项测试技术方面有建树者优先；

English gloss: Applicants with demonstrated achievements in specialized testing, including test automation, server performance testing, and security testing, are preferred.

Chinese source span and English gloss	Type	Chinese character interval
自动化、服务器性能、安全等专项测试技术 <i>English gloss: Specialized testing techniques including test automation, server performance testing, and security testing</i>	S	[1,20)

Rationale: retain 自动化、服务器性能、安全等专项测试技术 as one S. 测试技术 (testing techniques) is the necessary shared head. Security or server performance alone becomes a domain/property; do not insert testing words absent at those positions.

Case 143 Objects of professional consulting

Record ID: 1829-s0002. Related entry: [EX107](#).

Full Chinese source

3. 为客户提供海外购房法律法规、贷款政策、交房、租赁等环节的专业咨询（公司提供相关培训课程）；

English gloss: 3. Provide clients with professional advice on laws and regulations, loan policies, property handover, leasing, and other aspects of overseas home purchases (the company provides relevant training courses).

Chinese source span and English gloss	Type	Chinese character interval
<u>海外购房法律法规、贷款政策、交房、租赁等环节的专业咨询</u> <i>English gloss: Professional advice on laws and regulations, loan policies, property handover, leasing, and other aspects of overseas home purchases</i>	S	[8,35)

Rationale: retain the complete professional-consulting S. Laws and regulations and the other items share the consulting head as its objects, without nested K. Parenthetical company-provided training is a benefit and excluded.

Case 167 Shared action in a design list

Record ID: 1834-s0008. Related entry: [EX107](#).

Full Chinese source

3、具有 1 年以上智能家居、楼宇自控、综合布线、安防监控、家庭影院等设计经验；

English gloss: 3. Have one year or more of design experience in smart homes, building automation, structured cabling, security surveillance, home theatres, and related systems.

Chinese source span and English gloss	Type	Chinese character interval
<u>智能家居、楼宇自控、综合布线、安防监控、家庭影院等设计</u> <i>English gloss: Design of smart homes, building automation, structured cabling, security surveillance, home theatres, and related systems</i>	S	[8,35)

Rationale: the list shares design, so retain one complete S and exclude experience. Smart homes or home theatres alone loses the design activity; do not add words.

Case 194 Splitting spans under a shared experience requirement

Record ID: 1838-s0008. Related entry: [EX119](#).

Full Chinese source

4、三年以上大中型互联网公司经营或 BI 相关工作经验，互联网公司广告运营分析优先；

English gloss: 4. At least three years of relevant work experience in internet company operations or BI at medium-sized or large companies; experience in advertising operations analysis at internet companies preferred.

Chinese source span and English gloss	Type	Chinese character interval
<u>互联网公司经营</u> <i>English gloss: Internet company operations</i>	S	[9,16)
BI	S	[17,19)
<u>互联网公司广告运营分析</u> <i>English gloss: Advertising operations analysis at internet companies</i>	S	[26,37)

Rationale: current R23 selects 互联网公司经营 (internet company operations), BI, and 互联网公司广告运营分析 (advertising operations analysis at internet companies) as three S spans. Shared experience context qualifies the first two. BI 相关工作经验 (BI-related work experience) is continuous source text; selecting the shorter BI is a boundary choice, not an added-word issue.

Version difference: the earlier saved annotation version used 互联网公司经营或 BI 相关工作经验 at [9,25), S, and 互联网公司广告运营分析 at [26,37), S. The current handbook revises guidance only; it does not overwrite old data or change historical entity counts.

Part III Using conditional examples and pending records

The following notes distinguish decisions adopted in this edition from historical answers that still require review. Use confirmed items according to their case notes. Retain uncertainty notices for disputed items rather than treating them as the sole reference answers under revised rules.

- Python 语法与标准库 (Python syntax and the standard library): one K span in an established applicant knowledge requirement, retaining the shared Python qualifier. A textbook title alone is excluded. See EX056.
- 会使用 Excel (able to use Excel): the confirmed span is Excel, S. For 会修电脑 (can repair computers) and 会写 SQL (can write SQL), retain the specific action. Assess other sentences specifying a task individually; see EX039.
- 确保数据备份策略有效 (ensure the data-backup strategy is effective): no entity in this example, which states only a desired outcome. Assess other sentences by their content, not by ensure alone. See EX030.
- 英语沟通 (English communication): no fixed T. Natural-language expression ability is L; general communication/coordination is T. Request review when the full record does not establish the emphasis; see the teaching contrasts in the main text.
- Category terms and specific names: exclude a category that only introduces a list; independently assess any separate category-level requirement. See R21 and EX051.
- Shared wording: R14 and R23 permit splitting when meaning is preserved. A shared object does not always require merging, and a conjunction alone does not justify splitting.
- SQL 查库 (querying a database with SQL) is confirmed as S; see EX057. Academic research in case 158 retains its historical answer pending D06 and is not a confirmed reference answer under revised research criteria.
- Search-engine enthusiast/advanced-user identities, capability-development headings, and understanding business/customer needs/product-line responsibility: retain conditional or pending status when context is insufficient; do not invent definitive answers.
- Display shorthand does not authorize source changes. Earlier tables sometimes omit example spaces in Python 基础知识, Python 语法, and SQL 原理. Actual annotations always follow source slicing; do not delete source characters.

Add reviewed new cases in a subsequent revision, recording the applicable rule and revision date.

Sources and maintenance

The examples derive from the main-handbook snapshot checked on 2026-09-28; see the individual notes for later revisions. EX IDs are navigation aids. Original sentences remain unchanged. Entity tables and highlights show the answers adopted in this edition; historical answers still requiring review are marked beside the case. Changes to this handbook do not mean that historical data files have been updated.

- Current Chinese main handbook. [Chinese handbook in the repository](#)

EX IDs and internal navigation are retained. Line numbers under Source location refer to historical Markdown files. Appendix B describes supporting files and the access arrangements that remain to be verified.

Appendix B Version and maintenance records

The following preserves historical execution records for tracing the relevant batches. Use the main operational rules for annotation and the corresponding experiment records for the paper's experimental status.

Version comparison: early discussions used the ordering $L > S > K > T$. The current handbook instead determines scope, boundaries, and local type case by case.

Document history: on 2026-09-29, annotator feedback prompted clearer explanations and integration of the case appendix; the rule version remained B.sop_v4.2.14. Saved data were not rewritten. The historical 2351 records used silver_plus_v4212_rev2; archived files and checksums remain unchanged. Later wording updates do not change that execution record.

Maintenance terms: frozen means preserving an archived version. A hash is a checksum used to detect file-content changes. Verify files by version and checksum when reproducing results.

Historical assisted annotation and review

Some additions to R08–R13 arose from historical prompt review. The associated prompt was recorded as PROMPT_silver_plus_v4211.txt; it must not replace prompts saved for other batches. Trace V4.2.10, the earlier batches generated by GPT-6.0 under V4.2.9, and historical Gold/IAA through their respective records.

The historical model-review interface updated only gpt6_label, gpt6_why, model, and gpt6_remark. Other values, row counts, order, and IDs remained unchanged. Recruitment text was data to annotate and existing labels were review candidates. Validate structure, character positions, and semantic decisions separately; save label changes as a new version.

Version status: the 38-record batch was processed through review or authorized re-review and incorporated into a separately saved 100-record Silver-plus version; the original was retained. Other records follow their own review status.

1. The 38 records comprise 31 reviewed as a whole, two literature-supported conditional confirmations (cases 152 and 160), and five processed under the project lead's knowledge/application interpretation. These are assisted annotation and review records; independent blind-annotation statistics are reported separately. The execution rationale for stream-computing K in case 102 is retained.

2. The revised 100-record set contains 419 entities: K87, S255, T77, L0. Thirteen records changed labels or boundaries; all fields of 62 originally undisputed records remained unchanged; zero unresolved records remained. These counts describe version changes, not accuracy or agreement, which require separate evaluation.

Historical P2 configuration (short spans, no partial words, and a jieba-derived view): training silver labels train_lskt_v4_silver; test reference labels data/test_lskt_v4_cws_simhuman980_hybrid.jsonl (2,601 = 980 SimHuman rule_v4 + 1,621 SOP-CWS; the same record IDs as Gold v2, with predictions and reference labels both snapped to jieba boundaries).

The historical reference layer combines rule overlays and model annotations, including 980 rule-overlay sentences. It is stored separately from human Doccano Gold. gold_canonical_v2.jsonl retains its original version.

Historical P2 record: the main table used saved model outputs followed by jieba processing, not a fresh official gpt-4o call under the SOP. The SOP extraction prompt, PROMPT_gpt4o_sop_extract.txt, was used for the diagnostic table at that time. Its content is determined by the archived file; do not assume that it matches the current handbook.

Historical P2 results: JobBERT 3M v4+jieba typed exact F1 0.4331; frozen ChatGPT dump+jieba exact 0.2854 / relaxed 0.6249. ChatGPT 0.6365 on Gold v2 uses a different reference layer and is reported separately.

Supporting maintenance files: handbook_B_overlap_adjudication.md (review of overlapping annotations); handbook_B_citations.md (the relationship between concept sources and project rules); LSKT_V4_RULE_CHANGELOG.md (rule changes); and handbook_B_review_20260907.md (historical correction records). Additional examples in those files are not automatically included in this handbook's 124 EX entries. The filenames help locate materials; access links remain to be verified and the files are not necessarily all public.

Record revision and publication status separately for the Word edition, GitHub, Overleaf, and model prompts. Preserve historical originals, frozen labels, and model-execution files.

D10 Historical versions and dataset counts

PROMPT_silver_plus_v4211.txt and silver_plus_v4212_rev2, recorded for the 2,351-record batch, belong to different historical records and are not interchangeable. The handbook, prompt, labels, and evaluation protocol actually used for each batch still need to be matched to the archived files.

That version's ledger contains 2451 records. Of the remaining 29, secondary classification yielded 3 confirmed-empty and 26 difficult/undetermined records. The conservatively filtered 1424 records formed the then-current training-candidate pool: 520 from the difficult source queue and 904 from the easy queue, before development-set separation. These counts describe that data version only. That organization step launched no training and left historical results unchanged.

Historical revision note: v4.2.12 incorporated the review decisions adopted at that time, while the earlier 100 records retained their original model-call records. That handbook revision introduced no new training or testing; other P01–P10 cases follow their individual status records.

B.sop_v4.2.14 names a handbook series, not a guarantee that all later Word files are identical. The September 29 clarification draft, September 30 reader edition, October 2 manuscript-alignment edition, and October 3 integrated drafts are different document revisions within that series. The October 6–7 human-review drafts are later maintenance materials, not evidence that earlier annotation used the same content.

According to the author's clarification, the B.sop_v4.2.14 series supported later model-assisted human reference annotation, corrections, and case discussions. Exact links between batches and revised files remain to be verified. This does not establish that earlier bulk annotation used the same series.

Access to materials: see the repository entry under Sources and maintenance in Appendix A. The corresponding local snapshots, historical texts and labels, completed review records, and public links remain to be verified. This edition does not provide paths that only work on an author's computer.

Appendix C Annotation decisions procedures and reproduction

This appendix brings together annotation decisions, procedures, and reproduction guidance. Use confirmed cases as stated in their notes. Cases and materials still requiring review are identified separately. D IDs are retained to trace the discussion; D10 on historical versions and counts appears in Appendix B.

Part One Annotation decisions

D01 Context available for decisions

Locations: preface, R21, EX114, and the reading guidance for full cases.

Confirmed: use only the current sentence, including its component clauses. Do not consult neighboring sentences, the job title, or the full advertisement to supply missing meaning. Request review if evidence is insufficient. Historical materials remain unchanged; their labels are not automatically revised by this decision.

D02 Shortest complete spans and shared wording

Confirmed: select the shortest continuous span that fully expresses the requirement. In case 103, retain 产品的 (of products) as the occupational object in 产品的探索 (product exploration), S. In a knowledge requirement, Python 语法与标准库 (Python syntax and the standard library) is one K span with the shared qualifier. Usually exclude 会 / 能够 (can) and 熟练 (proficiently). Generic tool use may be represented by the tool name; retain necessary actions and objects for a specific task.

Do not split mechanically by length or conjunctions. If these criteria do not establish one boundary for a new sentence, record alternatives and reasons and request review. Individual preference is not a final standard.

D03 Status of multiple candidate answers

Confirmed decisions: 会使用 Excel (able to use Excel) yields Excel, S. Python 语法与标准库 (Python syntax and the standard library) is one K span in a knowledge requirement. In EX051, omit a category that only introduces a list. In EX057, select SQL 查库 (query a database with SQL), S; in EX030, 确保数据备份策略有效 (ensure the data-backup strategy is effective) yields no entity. Case 102 selects 计算机相关专业 (a major in computer science or a related field), K, and retains the Java and Spring boundaries. 翻译项目资源接口 (resource liaison for translation projects) in case 153 is S. 测试理念 (a testing mindset) in case 127 is T, expressing work-related awareness and attitudes in this sentence; the other entities remain unchanged.

Confirmed case decision: 消息队列框架 (message-queue framework) in case 102 retains S. Judge whether the current sentence emphasizes knowledge or practical use. 熟悉 (be familiar with) and 了解 (understand) are clues, not sufficient grounds for K or S by themselves.

D04 Evidence for classifying tool names as S

Use, installation, development, practical operating experience, or a tool list tied to an immediately adjacent occupational task can support S. Explicit requirements for course knowledge, principles, or fundamentals support K.

Request review when only the tool name, familiarity level, or job title is available and a use/knowledge requirement cannot be identified. A historical tool-stack interpretation is a case-specific decision, not a default S rule for new sentences.

D05 Natural-language proficiency and general communication

Natural-language proficiency is L; general communication and coordination are T.

Assess the sentence context. Do not decide from an assumed industry orientation or assign T to every occurrence of 英语沟通 (communication in English). If the emphasis remains unclear, request review rather than applying a fixed label priority. Historical labels are not retrospectively overwritten.

D06 Classifying research related expressions

Assess research-related expressions as T for general abilities, S for specific occupational research activities or methods, and K for knowledge requirements. 论文 / 实验 / 课题 (papers / experiments / research projects) are clues, not required words.

论文 (papers), 实验 (experiments), and 课题 (research projects) are only clues. Their absence does not rule out research activity, and their presence does not necessarily establish practical ability. Case 158 retains its historical T pending review before use under the current criteria. Existing agreement results do not validate this revision.

D07 Annotating categories alongside specific examples

Do not separately annotate a general category that only introduces a list. If the general item and the specific items each express an identifiable knowledge, skill, or work requirement in the current sentence, they may be annotated separately, provided their spans do not overlap. Non-overlapping text alone is not sufficient grounds for annotating both.

Confirmed case: 协助销售 (support sales) in case 197 describes sales-support work. Annotate it, 指导客户广告投放 (advise clients on ad placement), and 优化广告效果 (optimize advertising performance) separately as S.

D08 Occupational activities versus outcome targets

Determine whether the sentence assigns specific work or merely states a desired result. Do not decide mechanically from 优化 (optimize), 提高 (improve), 保证 (ensure), or similar verbs. In case 197, sales support, advertising guidance, and advertising optimization are work activities. 提升广告主的消耗以完成团队的销售目标 (increase advertisers' spending to meet the team's sales targets) states a performance goal and is not separately annotated. Request review if the distinction is unclear.

Part Two Annotation procedures and reproduction

D09 New annotation versus historical reproduction

For new annotation, record the handbook version and date used. To reproduce a historical experiment, use the prompts, labels, predictions, and evaluation rules saved for that batch. Do not replace its historical inputs with the current handbook.

Appendix B describes the handbook series used for later model assistance and the batch-to-file links still requiring verification. The current handbook does not replace the actual inputs to historical experiments.

D11 Difficult records and empty annotations

If it is unclear whether to annotate, which boundaries to select, or which type to assign, record the source sentence, candidate annotations, and question, and submit the record for review.

Confirmed empty means the sentence has been checked and contains no target entities. Pending review means that the answer has not been settled. Record these states separately. Provisional labels used for compatibility with an older workflow are not final answers and must not be used as confirmed labels for training or evaluation.

Manage difficult records and confirmed-empty records separately under R11 and R19. Compatibility arrangements for historical fields do not define new final labels.

D12 How cases were confirmed and how they may be used

Handbook examples explain the rules and support annotation decisions. Confirmed means that the handbook has adopted the case decision. It does not necessarily mean that the case was independently annotated blind or belongs to an independent test set.

Describe independent human annotation, human review after seeing model suggestions, and retention of historical answers according to the actual records. Clearly mark cases still awaiting confirmation; do not present them as the sole reference answers.

D13 Access to supporting materials

Appendix B describes supporting-material versions, files, and the status of access verification. Historical handbooks, source texts and labels, and review records each correspond to their own files and versions.

Provide accessible fixed-version links for public materials and state access conditions for restricted materials. Do not use paths that work only on an author's computer.

D14 Recording and checking character offsets

Select entity spans from the source without rewriting the sentence or adding absent words. Count Unicode code points from zero. Include the start position and exclude the end position. Slicing the source at that interval must return exactly the annotated entity text.

The annotation tool may record offsets, which should be checked after export. If its counting convention differs from the project's convention, normalize it during conversion. Investigate any mismatch between text and offsets before correcting the record. Preserve historical archives unchanged.

Part Three Reading guidance and maintenance

D15 Consistency between rules and examples

When a rule or case decision changes, check the main explanation, source highlighting, entity table, character intervals, and Chinese and English editions together.

Keep the original case sentence unchanged. Record changes to entity boundaries or types, distinguishing the current illustrative answer from historical annotations. Mark any unconfirmed treatment beside the case.

D16 Unclear source wording

Record spelling errors, incomplete expressions, and unclear wording in the notes. Do not silently rewrite the source.

If the current sentence still clearly identifies an entity, annotate the original wording. If scope, boundaries, or type remain uncertain, request review rather than guessing. Do not annotate passages confirmed to be out of scope. Correct existing misannotations in a revised copy and retain a change record.

D17 Current rules and historical records

The main text explains current annotation rules. Historical handbooks, prompts, batch information, and change records belong in Appendix B or the supporting archive.

State the date, version, and purpose of historical materials so that earlier plans or procedures are not mistaken for current instructions. The public handbook should retain information needed to understand the rules and reproduce the work.

D18 Concept sources and project rules

External literature provides concepts and annotation-method guidance. Whether to extract a Chinese passage, where to place its boundaries, and which final type to assign are determined by this handbook and the current sentence.

Identify project-specific decisions as such rather than presenting them as direct prescriptions from external literature for the Chinese passage.

References

Reference markers correspond to the entries below. [Project rule] indicates an annotation rule established for this project, not an external reference. References inform concepts, task design, and annotation methods. Unless explicitly stated otherwise, exact Chinese boundaries and final labels are project decisions.

1. [ESCO14] le Vrang, M., Papantoniou, A., Pauwels, E., Fannes, P., Vandensteen, D., and De Smedt, J. 2014. ESCO: Boosting job matching in Europe with semantic interoperability. *Computer* 47(10):57–64. DOI: 10.1109/MC.2014.283. Portal: skill_main (https://esco.ec.europa.eu/en/classification/skill_main); Skills pillar (<https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/skills-pillar>); Knowledge (<https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/knowledge>); Skill (<https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/skill>).
2. [EQF] Council of the European Union. 2017. Council Recommendation of 22 May 2017 on the European Qualifications Framework for lifelong learning (2017/C 189/03). Official Journal of the European Union C 189:15–28. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32017H0615%2801%29>
3. [ESCO-L] European Commission. ESCO skills pillar, class *Language skills and knowledge* (hierarchy letter L). https://esco.ec.europa.eu/en/classification/skill_main ; <https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/skills-pillar>
4. [ESCO-T] European Commission / Cedefop. 2022. Towards a structured and consistent terminology on transversal skills and competences. ESCO/EQF expert group final report. <https://esco.ec.europa.eu/en/about-esco/publications/publication/towards-structured-and-consistent-terminology-transversal> ; report facts concept: <http://data.europa.eu/esco/skill/be6ab363-3de1-427f-a8ef-85d5b0250822>
5. [ESCO-Q] European Commission. ESCO: Qualifications. <https://esco.ec.europa.eu/en/classification/qualifications>
6. [Z22] Zhang, M., Jensen, K. N., Sonniks, S., and Plank, B. 2022. SkillSpan: Hard and Soft Skill Extraction from English Job Postings. In *NAACL-HLT*, 4962–4984. DOI: 10.18653/v1/2022.naacl-main.366. <https://aclanthology.org/2022.naacl-main.366/> (Annotation guidelines: Appendix B of that paper; education example: B.2, item 8.)
7. [Say18] Sayfullina, L., Malmi, E., and Kannala, J. 2018. Learning Representations for Soft Skill Matching. In *Analysis of Images, Social Networks and Texts* (AIST 2018), LNCS 11179, 141–152. https://doi.org/10.1007/978-3-030-11027-7_15 ; arXiv:1807.07741.
8. [AP08] Artstein, R., and Poesio, M. 2008. Inter-Coder Agreement for Computational Linguistics. *Computational Linguistics* 34(4):555–596. <https://aclanthology.org/J08-4004/>
9. [Kr95] Krippendorff, K. 1995. On the reliability of unitizing contiguous data. *Sociological Methodology* 25:47–76.
10. [TKS02] Tjong Kim Sang, E. F. 2002. Introduction to the CoNLL-2002 Shared Task: Language-Independent Named Entity Recognition. In *CoNLL*. <https://aclanthology.org/W02-2024/>
11. [FM09] Finkel, J. R., and Manning, C. D. 2009. Nested Named Entity Recognition. In *EMNLP*, 141–150. <https://aclanthology.org/D09-1015/>
12. [Yu20] Yu, J., Bohnet, B., and Poesio, M. 2020. Named Entity Recognition as Dependency Parsing. In *ACL*, 6470–6476. <https://aclanthology.org/2020.acl-main.577/>
13. [UD20] Nivre, J., de Marneffe, M.-C., Ginter, F., Hajič, J., Manning, C. D., Pyysalo, S., Schuster, S., Tyers, F., and Zeman, D. 2020. Universal Dependencies v2: An Evergrowing Multilingual Treebank Collection. In *LREC*, 4034–4043. <https://aclanthology.org/2020.lrec-1.497/> ; Chinese ellipsis: <https://universaldependencies.org/zh/dep/orphan.html>

14. [D18] Nakayama, H., Kubo, T., Kamura, J., Taniguchi, Y., and Liang, X. 2018. doccano. <https://github.com/doccano/doccano>
15. [ONET] Peterson, N. G., Mumford, M. D., Borman, W. C., Jeanneret, P. R., Fleishman, E. A., Levin, K. Y., Campion, M. A., Mayfield, M. S., Morgeson, F. P., Pearlman, K., Gowing, M. K., Lancaster, A. R., Silver, M. B., and Dye, D. M. 2001. Understanding Work Using the Occupational Information Network (ONET). *Personnel Psychology* 54(2):451–477.
16. [Nav09] Navigli, R. 2009. Word Sense Disambiguation: A Survey. *ACM Computing Surveys* 41(2):1–69.
17. [PB05] Palmer, M., Gildea, D., and Kingsbury, P. 2005. The Proposition Bank: An Annotated Corpus of Semantic Roles. *Computational Linguistics* 31(1):71–106.
18. [LK77] Landis, J. R., and Koch, G. G. 1977. The Measurement of Observer Agreement for Categorical Data. *Biometrics* 33(1):159–174. Used to interpret agreement measures; high agreement alone does not establish label correctness.
19. [ESCO-C] European Commission. ESCO: Competence. <https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/competence>
20. [ESCO-R] European Commission. ESCO: Skill reusability level. <https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/skill-reusability-level>
21. [ESCO-TG] ESCO expert group. Report of the expert group on transversal skills and competences, §5.3. <https://esco.ec.europa.eu/system/files/2022-05/MSWG%2014-04%20Report%20of%20the%20expert%20group%20on%20transversal%20skills%20and%20competences.pdf>